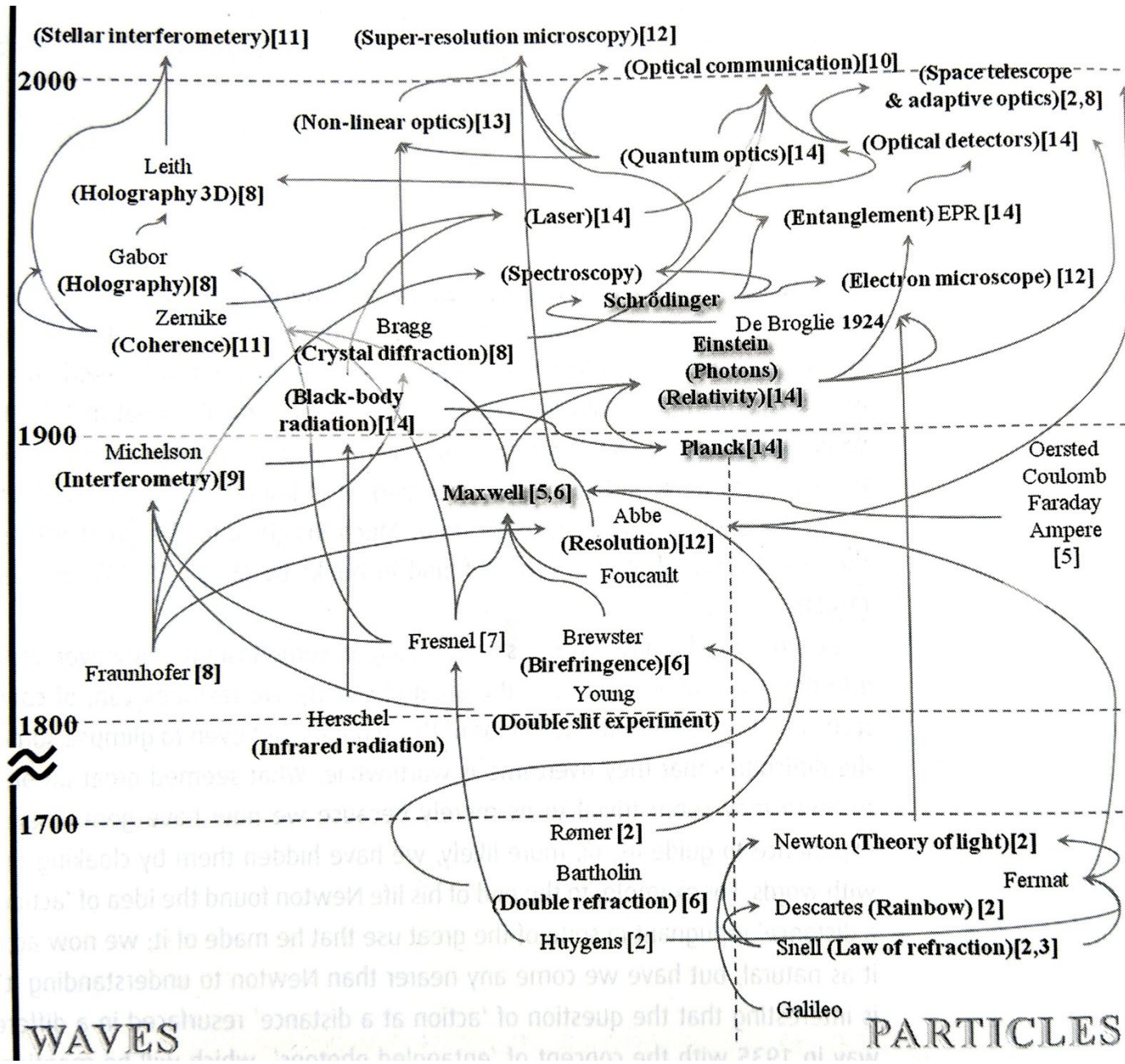
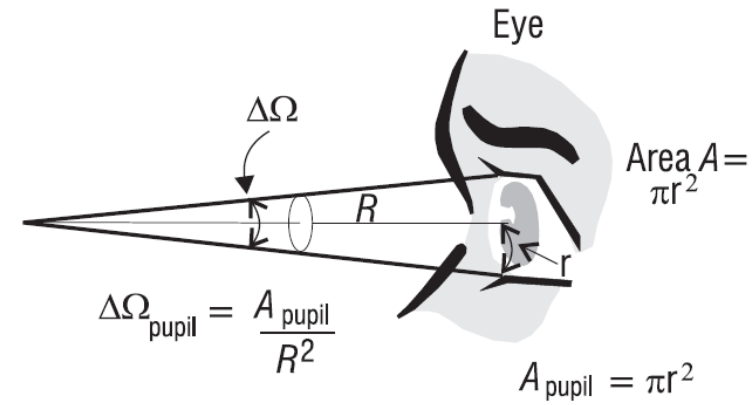
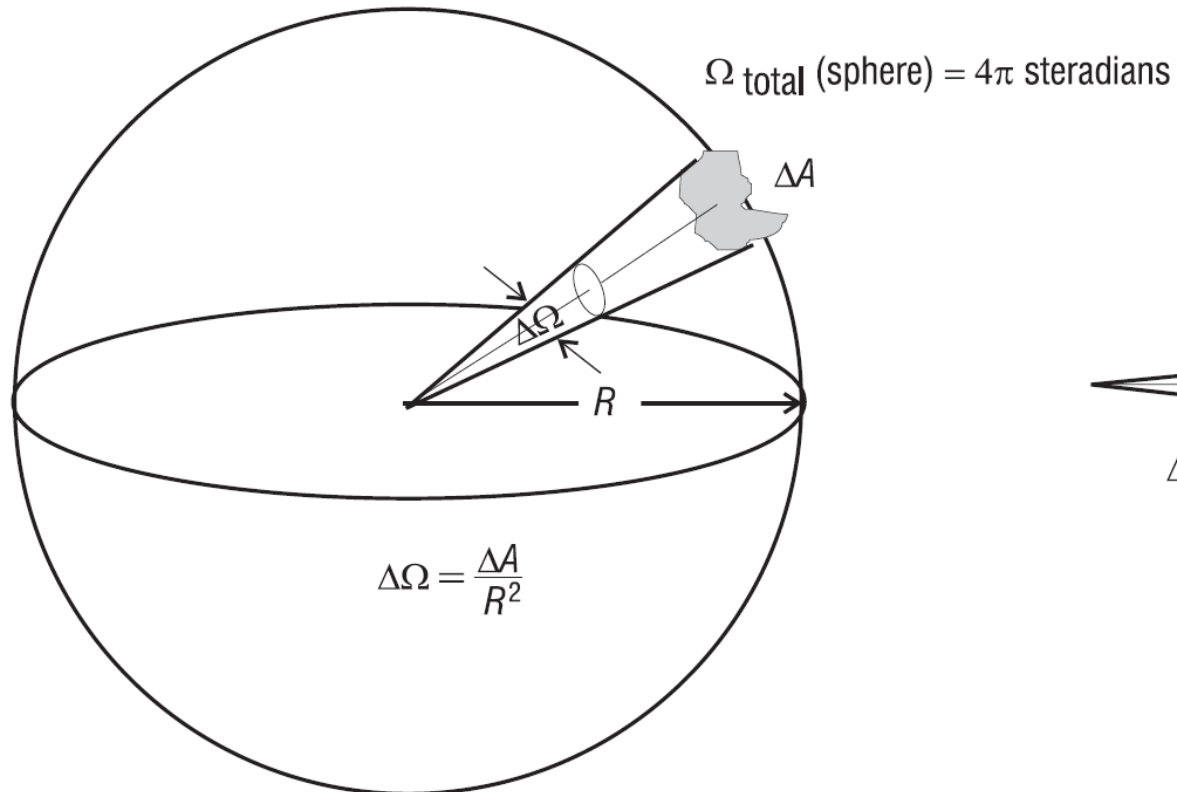
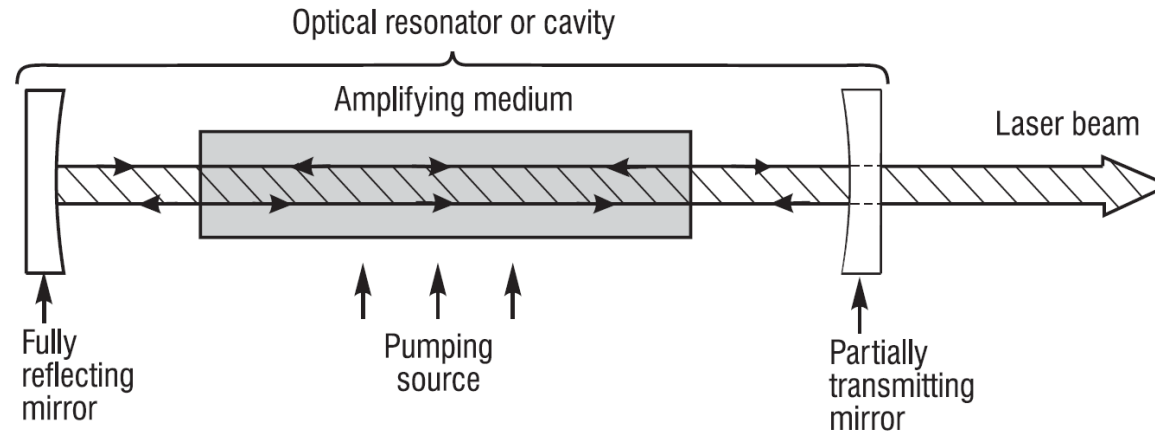


Historical Connections



“Laser Match”



Q: “Laser Match”

By what factor would the light intensity of a lit match (1 m away) be increased if all of its light could be concentrated into a cone just narrow enough to enter the pupil (2 mm dia.) of the observer's eye?

- A. $2 \times$
- B. $80 \times$
- C. $500 \times$
- D. $30000 \times$
-  E. $4000000 \times$

Q: Sun Focused onto Retina

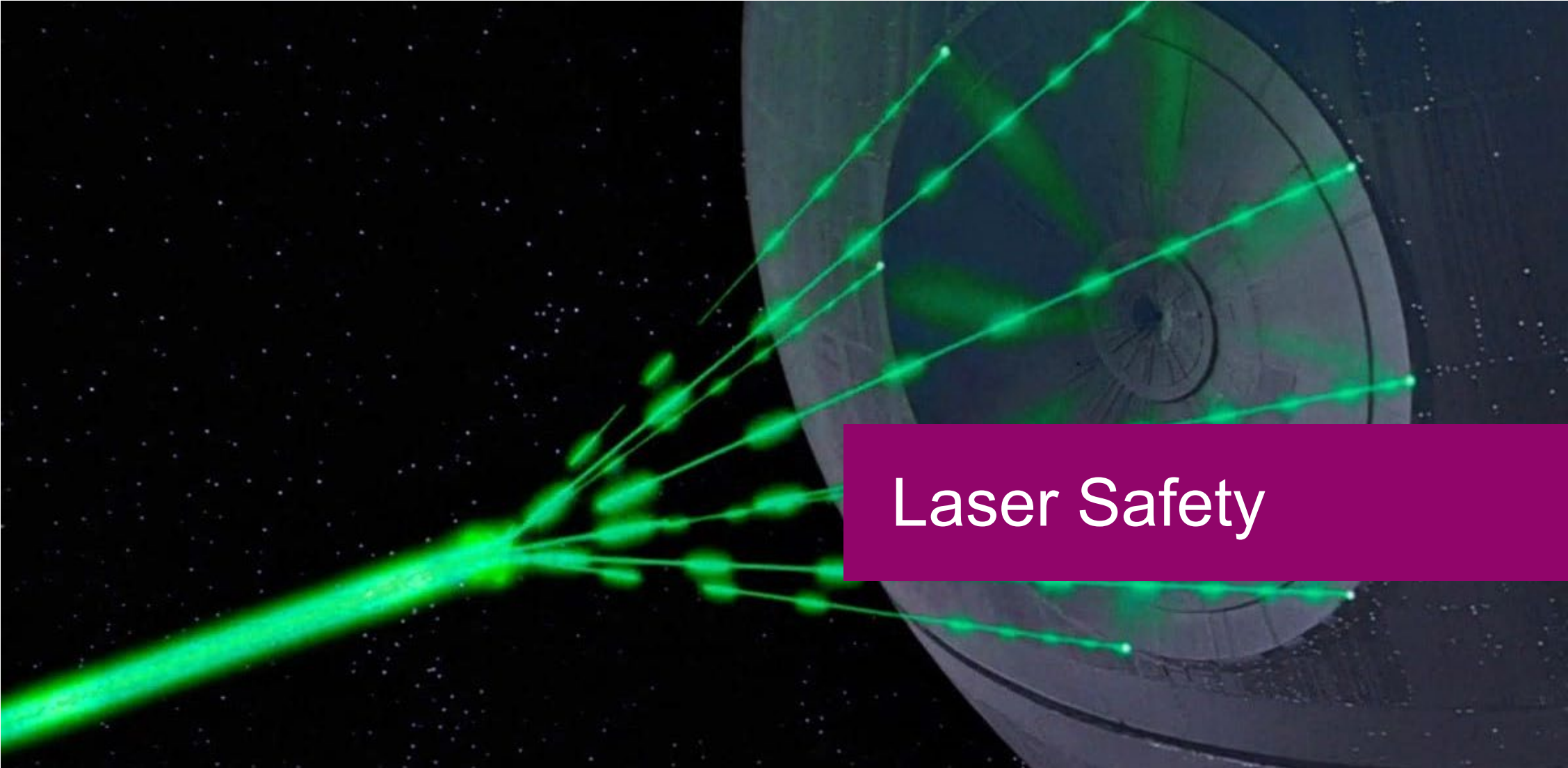
What is the intensity, in W/cm^2 , of the image of the Sun on the retina, as per Problem 2.11(a) in *Peatross & Ware*?

- A. 0.0014
- B. 0.014
- C. 0.14
- D. 1.4
- E. 14
- F. 140
- G. 1400
- H. 14000

Q: Laser Pointer Focused onto Retina

What is the intensity, in W/cm^2 , of the image of the 1 mW HeNe laser, as per Problem 2.11(b) in *Peatross & Ware*?

- A. 0.0039
- B. 0.039
- C. 0.39
- D. 3.9
- E. 39
- F. 390
- G. 3900
- H. 39000

A green laser beam enters from the bottom left, hits a circular mirror, and reflects in several directions. The background is a dark space filled with stars.

Laser Safety

Light and Lasers

Light exists all around us;

light bulbs, sunlight, torches, fire, lightning, etc.

It is a **mixture of different wavelengths** and is spatially incoherent.

Light radiates in all directions and is **not bundled**.

Normally, such light is **not a danger** for our eyes or skin.



Laser light is not a natural light. It is produced **artificially**.

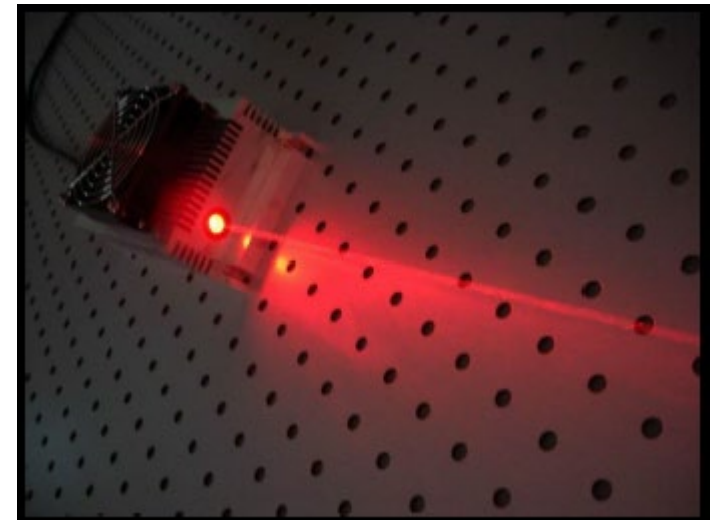
It consists of **monochromatic** light of a single wavelength.

It is spatially coherent in time and space,

and **bundled into a beam** which radiates in one direction.

Laser light has a **high density** and can contain a

high level of energy! It can be **precisely focused** !

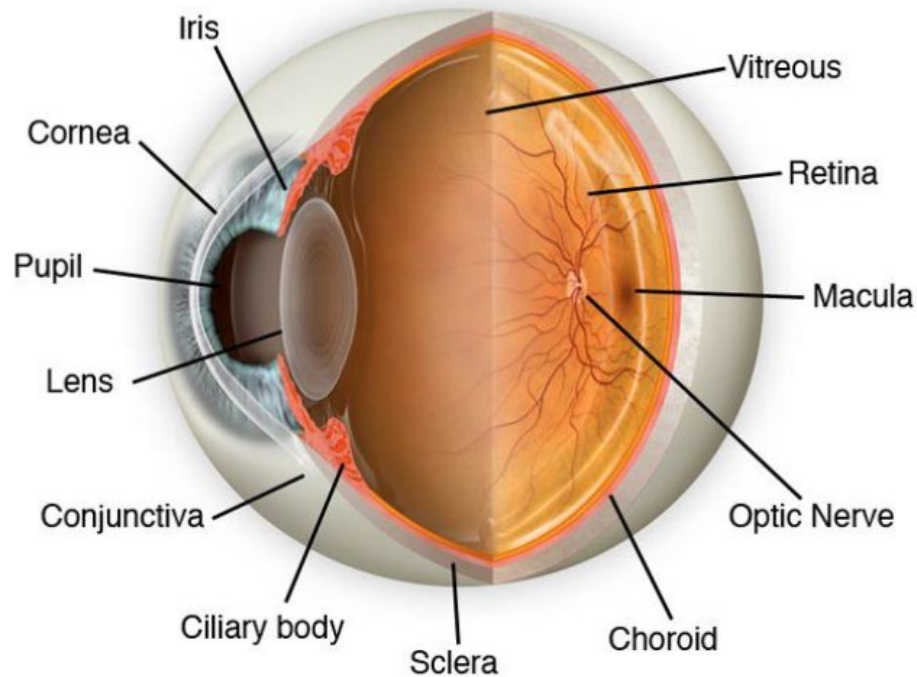


Some Basics of Lasers

- **Laser** = **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation
- Working modes:
 - CW laser = continuous wave:
 - produces a **continuous** beam of light with the same power over **a period longer than 0.25s**.
 - Pulsed laser:
 - A pulse laser produces a single or several pulses of light (each $< 0.25\text{s}$).
 - The pulse duration and repetition time depends on the design of the laser.
 - The pulse peak power can be very high!
- A very good rule when working with laser beams:

Never stare into the beam!

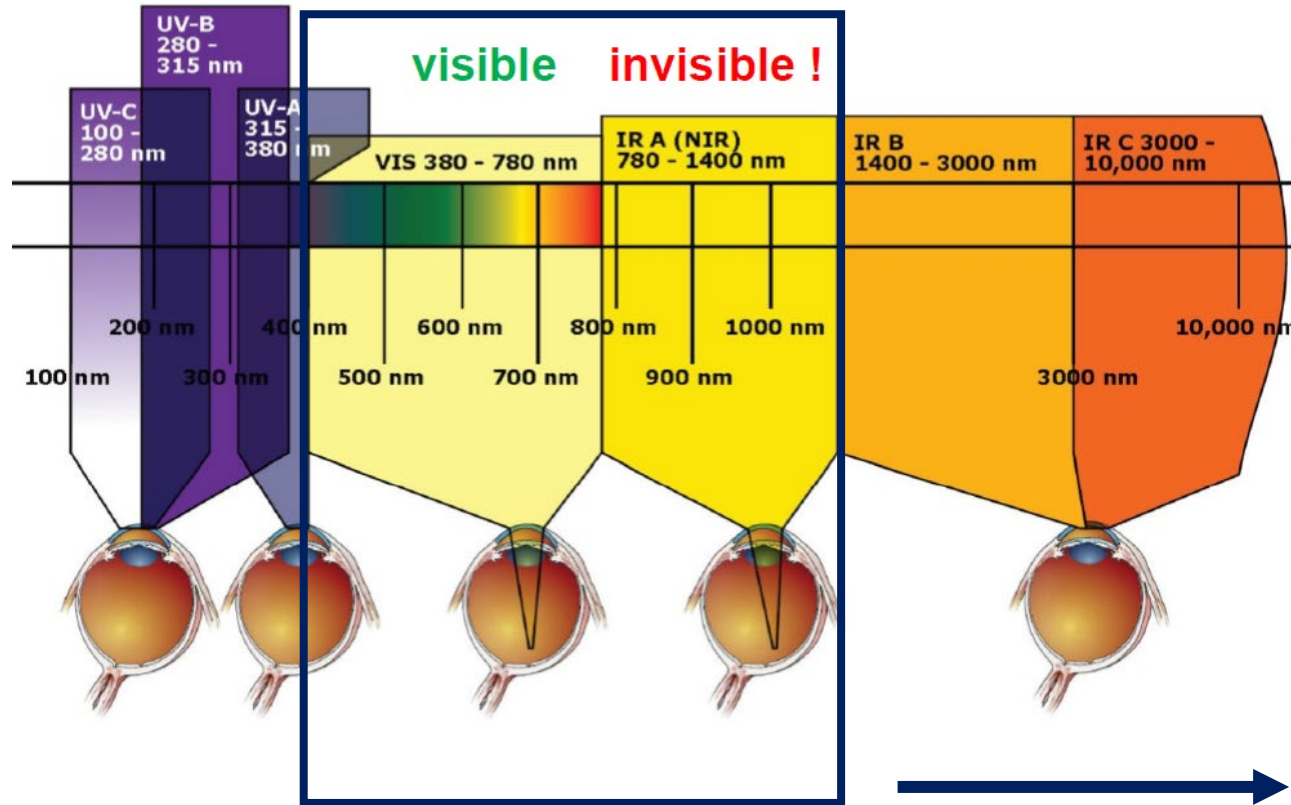
The Human Eye



- The human eye has **two self-protection** mechanisms:
 - **The pupil** limits the amount of light entering the eye depending on the brightness (density) of the light.
 - Note: The opening of the pupil for laser-related calculations and classification is defined as 7mm in diameter (or 0.39cm²).
 - **The eyelid** closes over the eye when a potential threat is recognised.
 - The reaction time to close the eyelid is defined as 0.25 s
 - (although more than 70% of all humans have a slower eyelid blink reflex!)
- Neither mechanism protects your eyes from a strong laser !

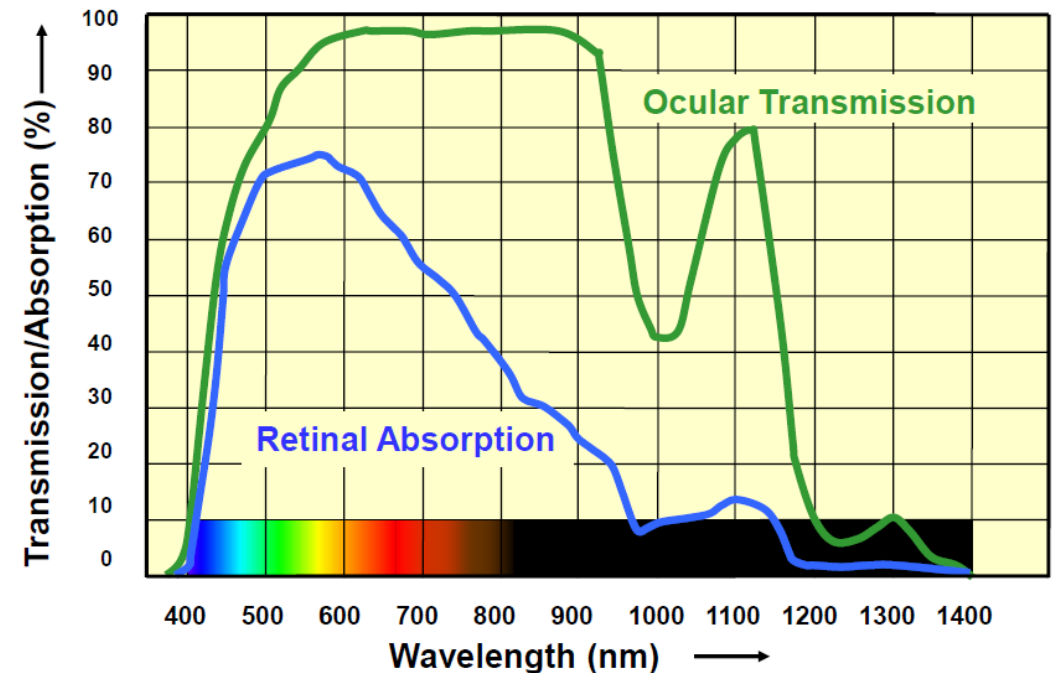
The Eye and Different Wavelengths

Light transmission into the human eye



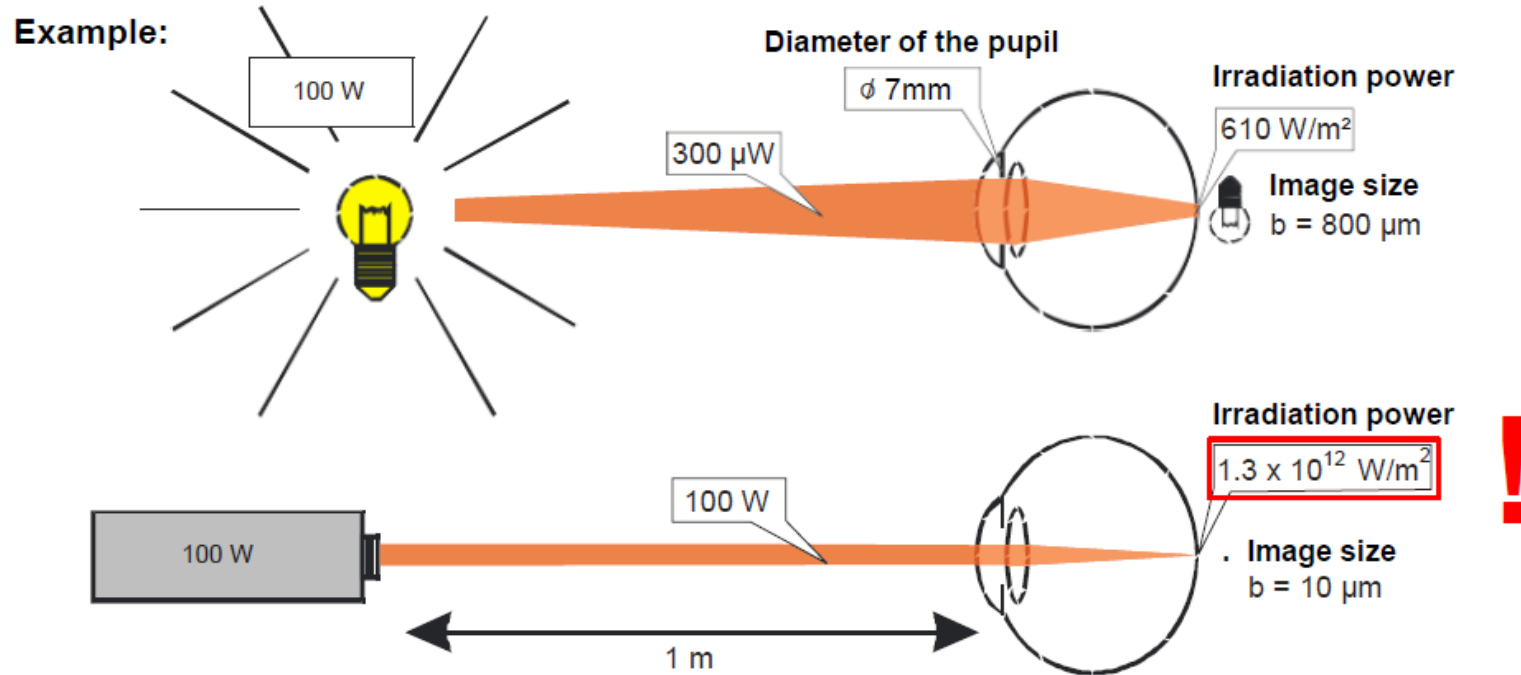
- 400nm –1400nm is the **retinal hazard region**.
- Outside this region: potential damage to cornea and lens

- The **visible light spectrum** that can be seen by a human eye is technically defined as 400nm – 700nm
- **However**, the human eye can receive a **wider spectrum** of light in the range of 400nm –1400nm which enters the eye



Another Example: Why are Light Bulbs Safe?

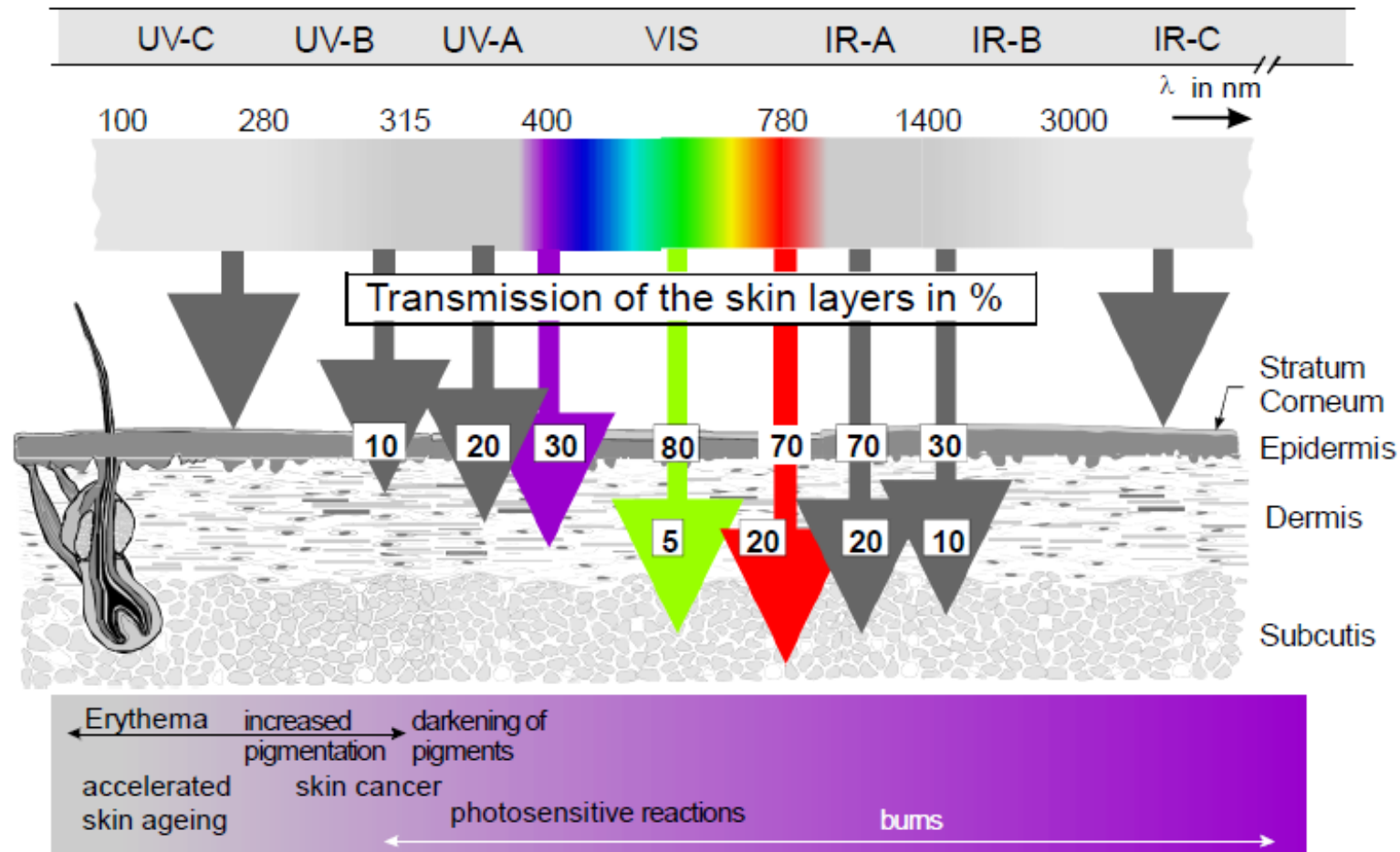
Density in W/m^2 : comparison between a light bulb and laser



Ratio of the irradiation powers:
$$\frac{E(\text{Laser } 100\text{ W})}{E(\text{Light bulb } 100\text{ W})} = 2.1 \times 10^9$$

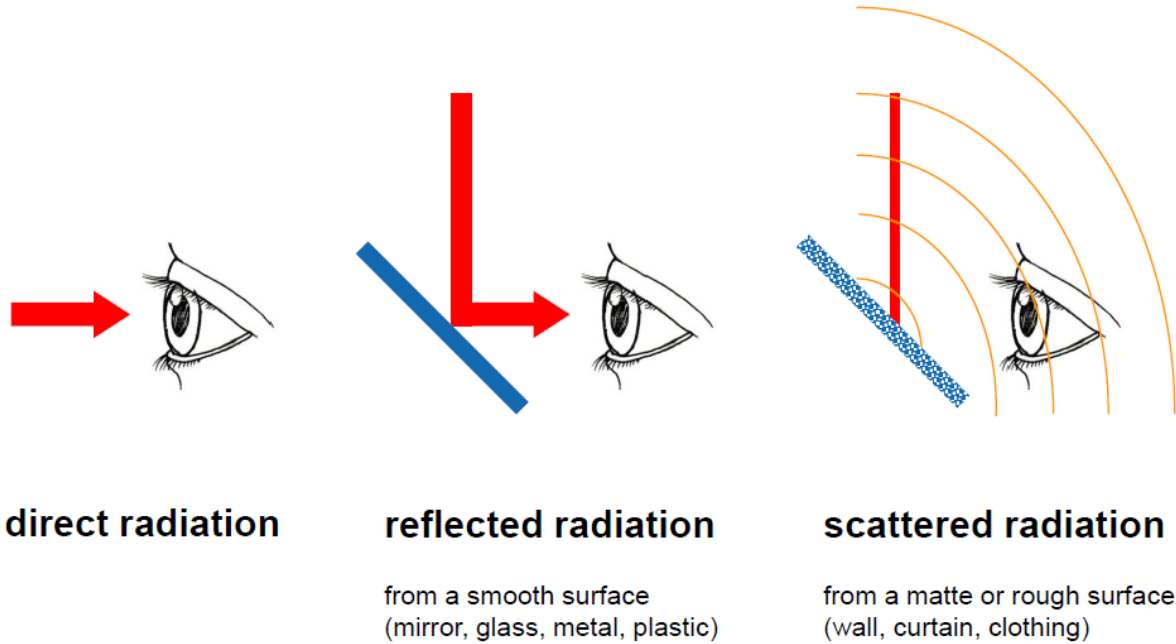
Further health hazards

Effects on the skin: transmission of the skin layers in %



Possible Eye Contact with Laser Beams

- Pathway of radiation

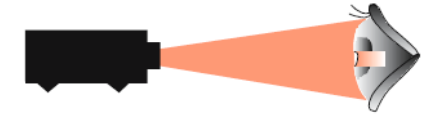


- Property of laser beam

widened laser beam

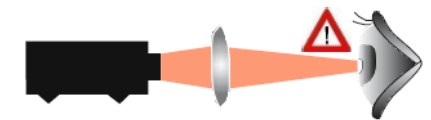
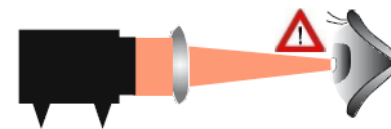


divergent laser beam



Laser light entering the eye from a widened or divergent beam, is limited by the pupil opening (7mm). Divergent lasers have less energy for a given area over increasing distances.

Keep your distance from divergent beams, especially open fibres !



Optical instruments focus these beams and increase the hazards of such lasers dramatically ! Do not use optical instruments ! This is important for class 1M and 2M lasers !

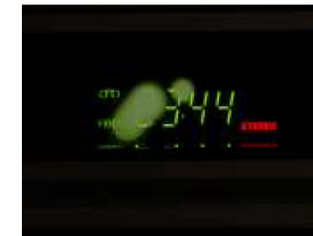
Possibly Occurring Damage

Laser effects on the eye and skin

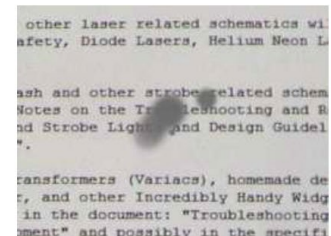
Wavelength	Spectrum	Effect on the eye	Effect on the skin
100 - 315 nm	UV	keratitis	sunburn, skin ageing, skin cancer
315 - 400 nm	UV	cataract	increased pigmentation, skin cancer
400 - 700 nm	VIS	retina injury	darkening of pigments, burn
700 - 1400 nm	IR	cataract, retina injury	burn
1400 - 3000 nm	IR	cataract, cornea burn	burn
3000 nm - 1mm	IR	cornea burn	burn

- ...We spare you the not so nice images of how these damages actually look like...

persisting spot at night



and on a day:



Classifications of a Laser

Class	Concept	Comment	Wave-length	Max. Power (cw)
1	The radiation emitted by this laser is not dangerous.	No need for protection equipment.	all	depending on wavelength, visible radiation $\ll 1\text{mW}$
1M	Not dangerous when used without optical instruments; may not be safe when optical instruments are used.	No need for protection equipment, if used without optical instruments.	200-4000nm	depending on wavelength, visible radiation $\ll 1\text{mW}$
2	Not dangerous due to the aversion response including the blink reflex.	No need for protection equipment.	400-700nm	1 mW
2M	The light that can hit the eye has the value of a class 2 laser. Depending on a divergent or widened beam, it may not be safe when optical instruments are used.	No need for protection equipment, if used without optical instruments.	400-700nm	Same as Class 2, distinction with measurement requirements
3R	The radiation from this laser exceeds the MPE values. The radiation is max. 5 x the class limit of class 1 (invisible) or 5 x the class limit of class 2 (visible). The risk is slightly lower than that of class 3B.	Dangerous to the eyes, safety glasses are recommended.	200-10'000nm	5 times the limit of Class 1 in UV and IR, and 5 times the limit for Class 2 in visible, i.e. 5 mW
3B	Old class 3B without 3R. The view into the laser is dangerous. Diffuse reflections are not regarded as dangerous.	Dangerous to the eyes, safety glasses are obligatory.	all	500 mW
4	Old class 4. Even scattered radiation can be dangerous, also danger of fire and danger to the skin.	Personal safety equipment is necessary (glasses, screens).	all	No limit

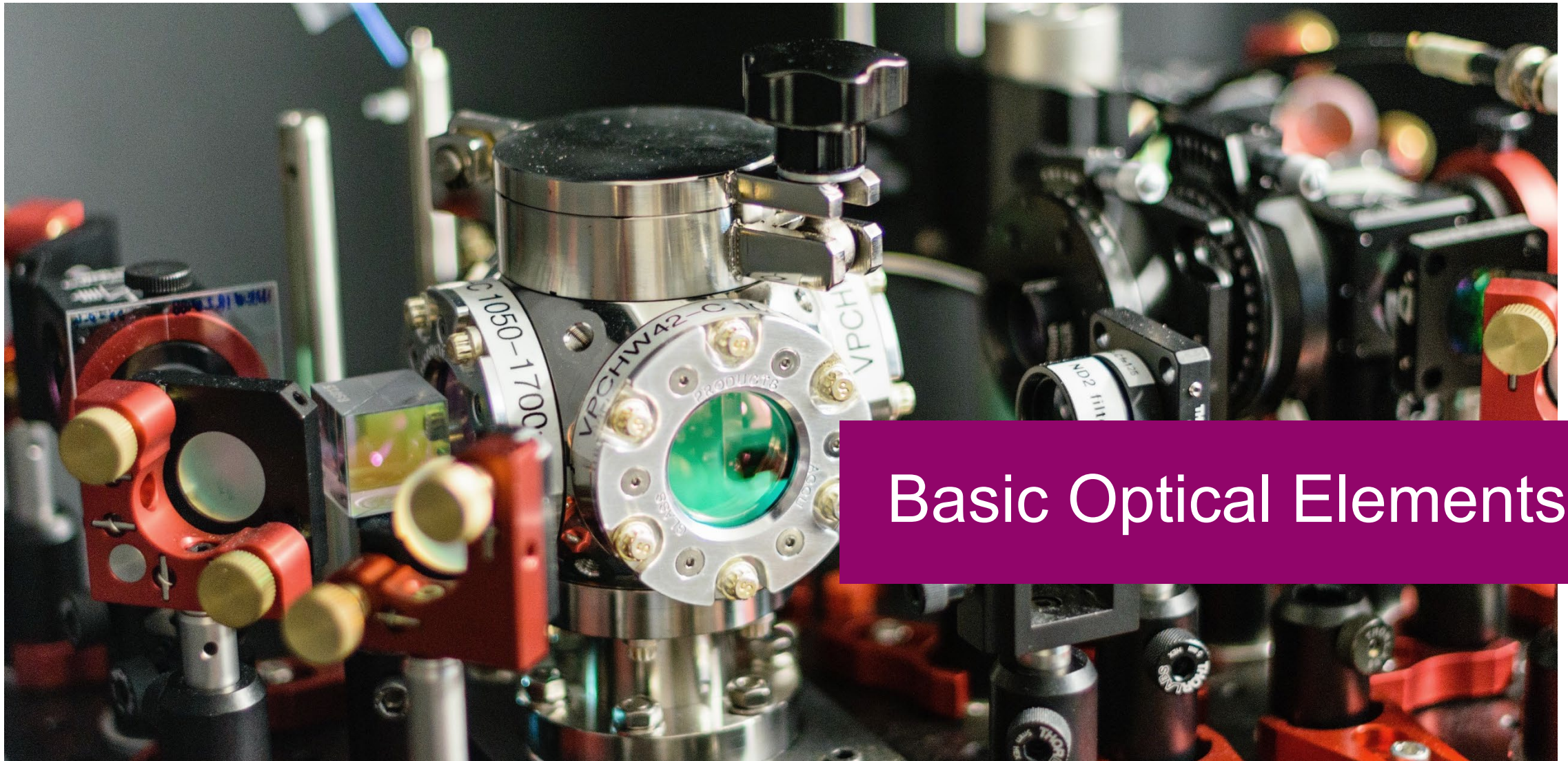
Classifications of a Laser

The maximum power of a class 1 laser depends on the wavelength.

Examples:

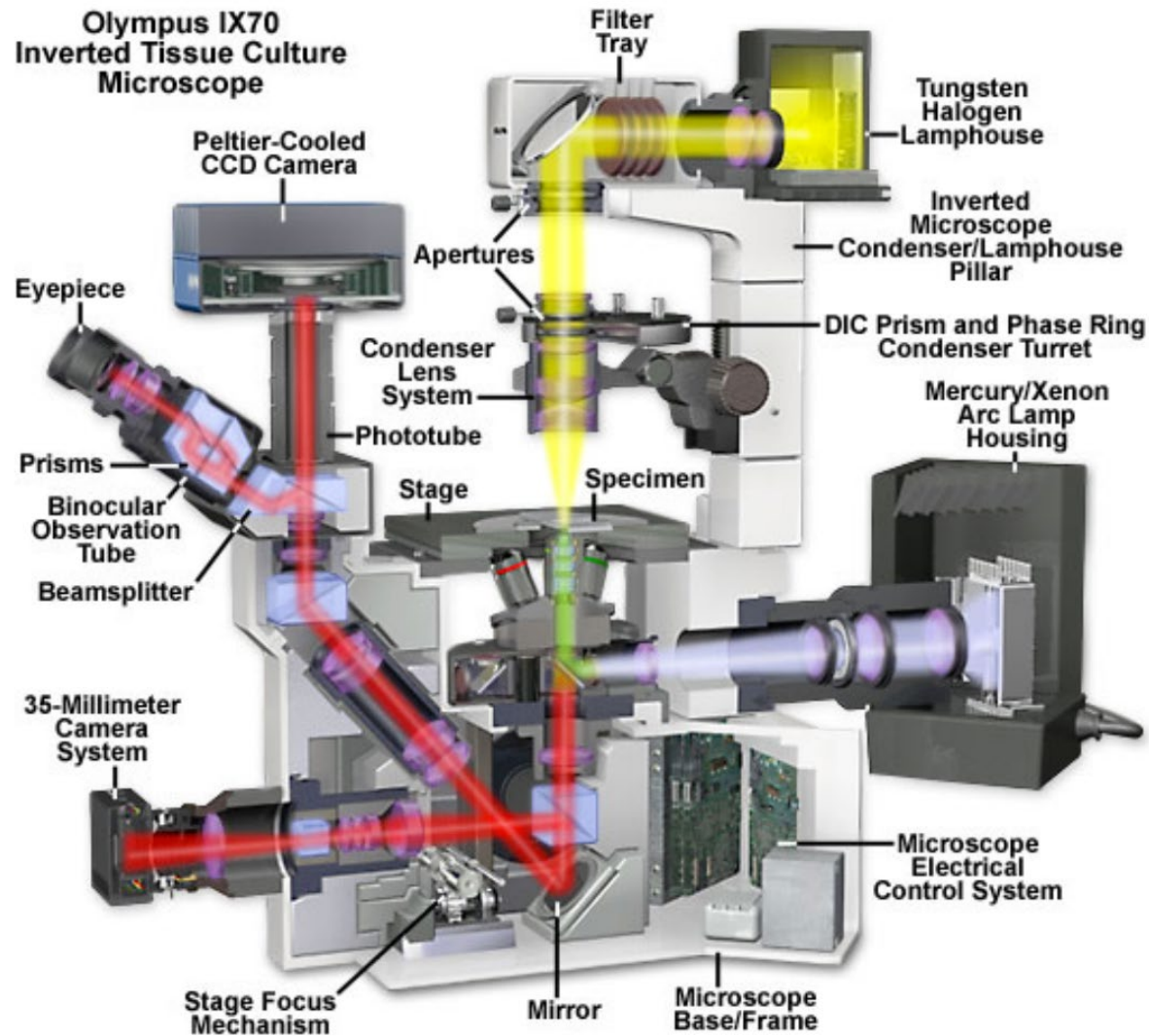
- UV = 7.9 μW
- Blue = 40 μW
- Yellow-Green = 400 μW

WARNING! If you dismantle a device, disconnect a fiber or remove casings or protection lids, the radiating laser power can suddenly be much higher than classified !



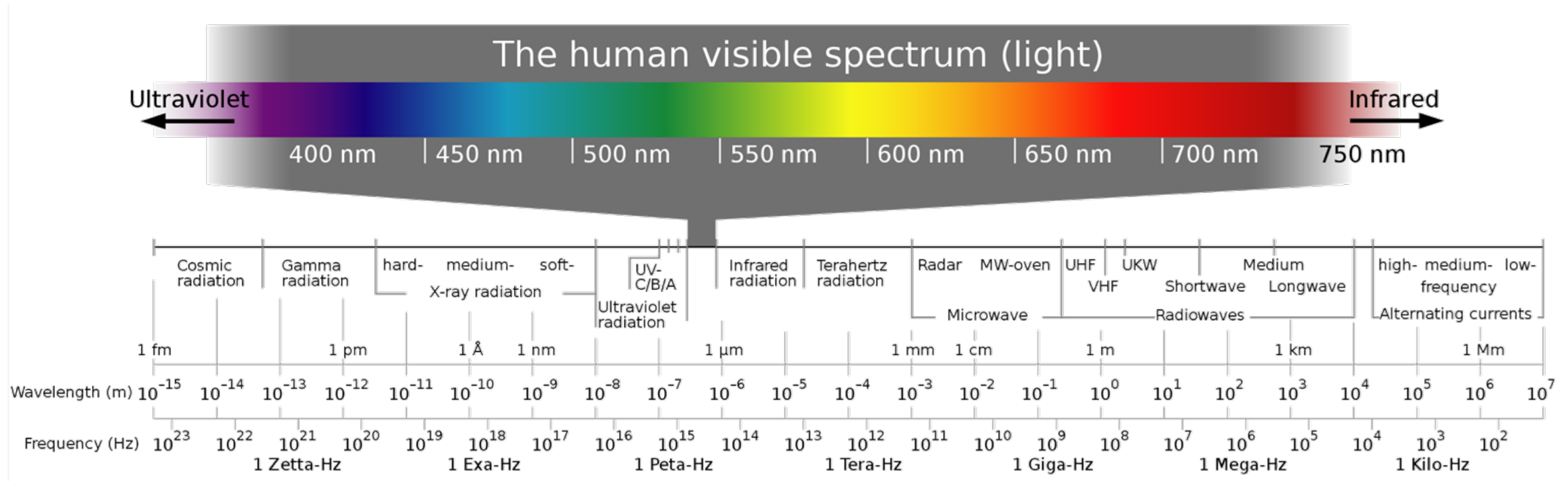
Basic Optical Elements

A commercial microscope



- Lenses
 - Mirrors
 - Polarizers
 - Beamsplitters
 - Filters
-
- Light sources
 - Incandescent, LED, Lasers
 - Detectors
 - Eyes, Semiconductor Silicon

Visible light



[wikimedia.org]

Ray Optics - Snell's Law

Ray Optics valid for large structures ($\gg \lambda$)

Light propagates along rays, with the speed of light in media with refractive index n slower than the speed of light in vacuum:

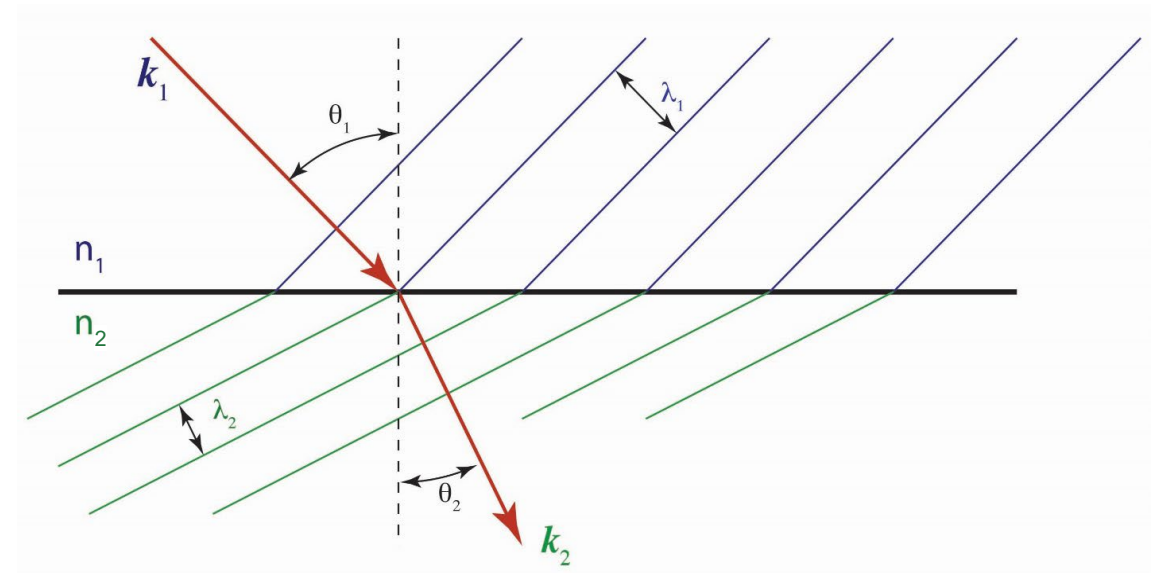
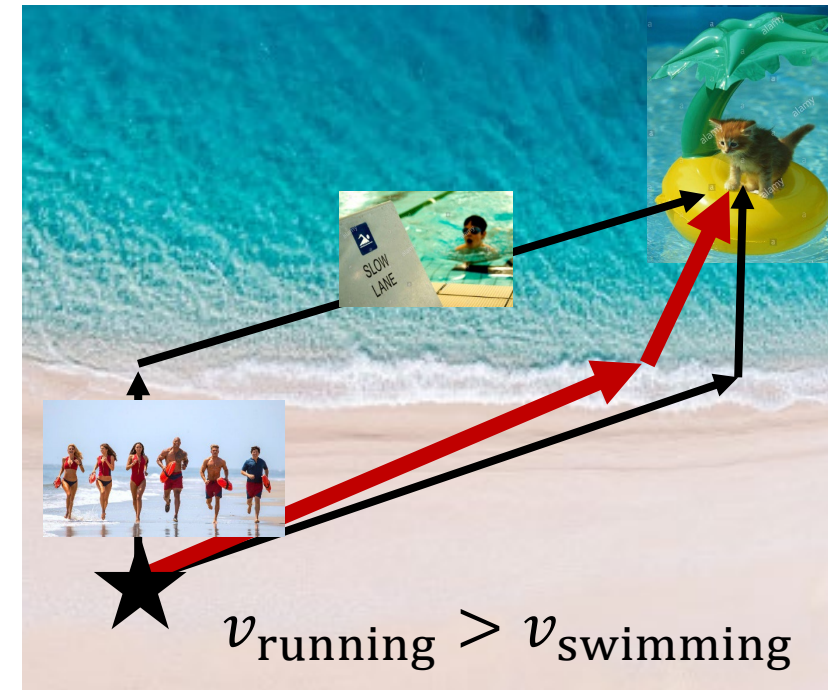
$$v_n = \frac{c_0}{n}$$

Fermat's Principle, 1662

Optical rays travel along the path of least time

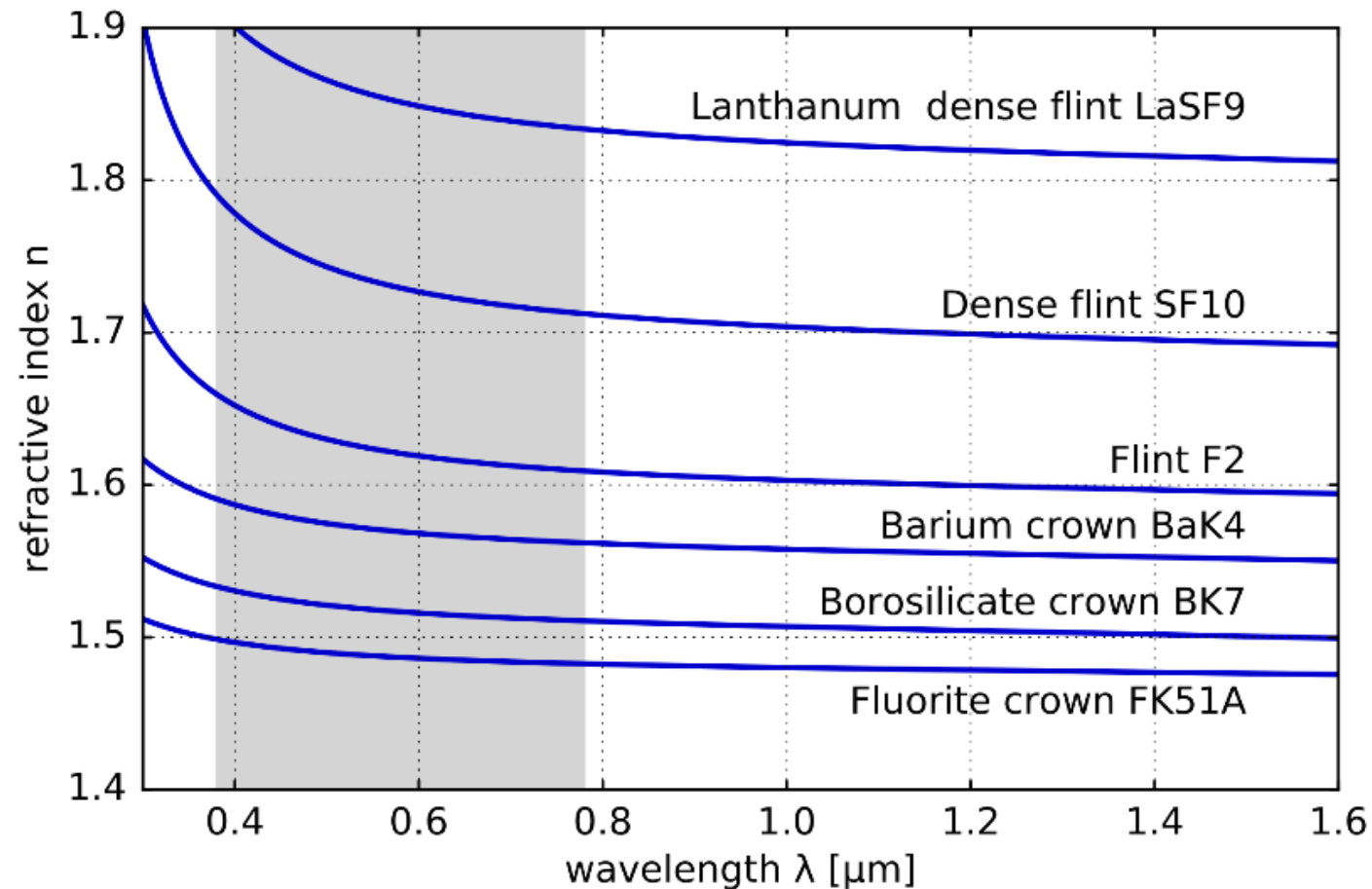
Snell's law

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$



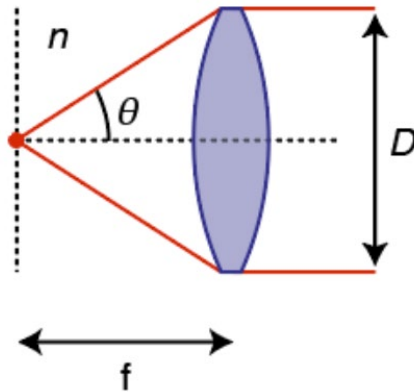
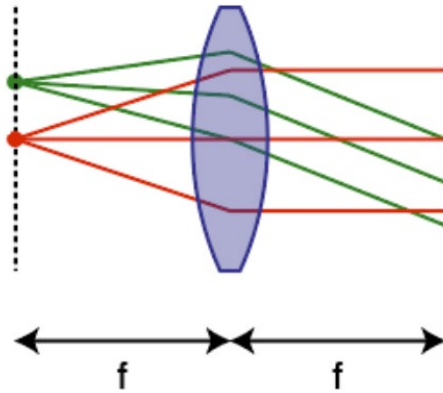
Lenses – Optical Materials

Large variation of refractive index and dispersion possible



<https://www.edmundoptics.de/knowledge-center/application-notes/optics/eo-preferred-glass-types/>

Ideal Lenses – Characteristics



- Perfect lens focuses all incoming parallel light rays to a point
- Conversely, all light rays coming from the same point in the focus plane are parallel after the lens
- Fourier transform
 - Lens converts position to angles and in reverse
- Focal length f
- Numerical aperture

Lenses – Spherical and Aspherical

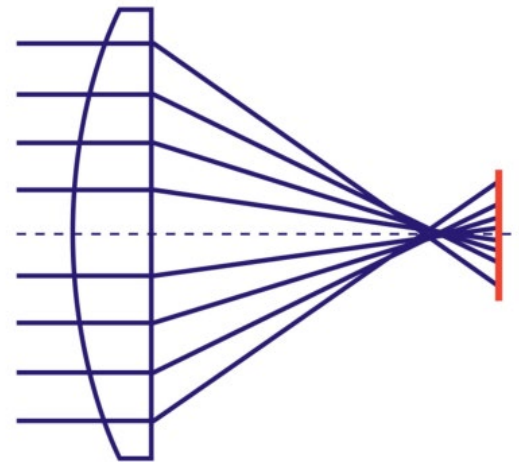
Spherical lenses

- easiest to fabricate
- The plano-convex lens is the most simple lens shape.
- Suffers from spherical aberrations.
- Spherical aberrations are strong for short focal lengths and rays far away from the optical axis

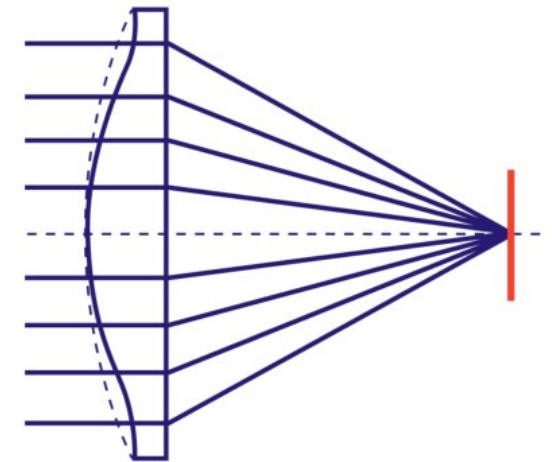
Aspheric lenses

- Lens surface different from sphere to correct spherical aberrations
- Much more difficult to fabricate
→ expensive

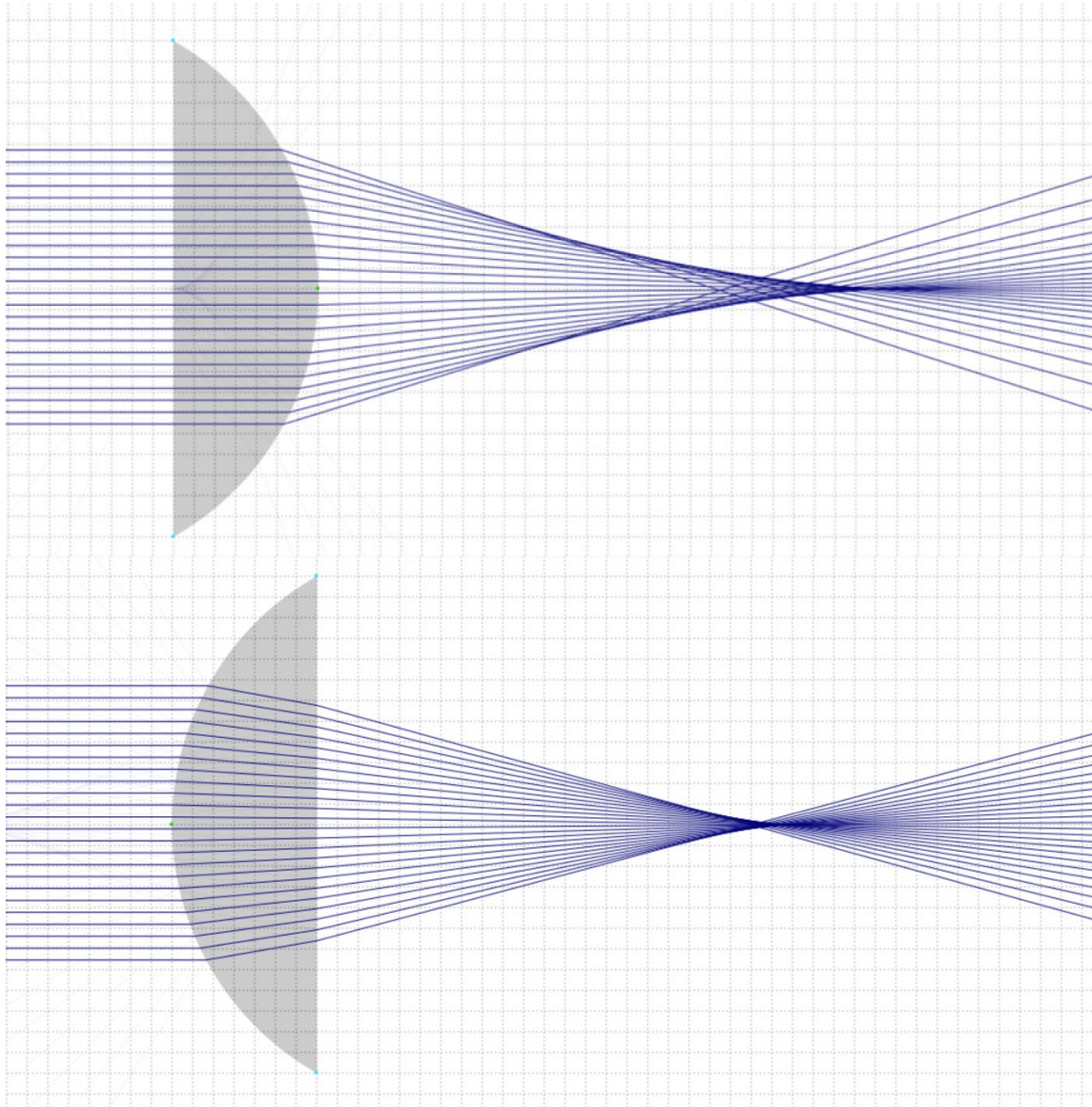
Spherical Lens



Aspherical Lens

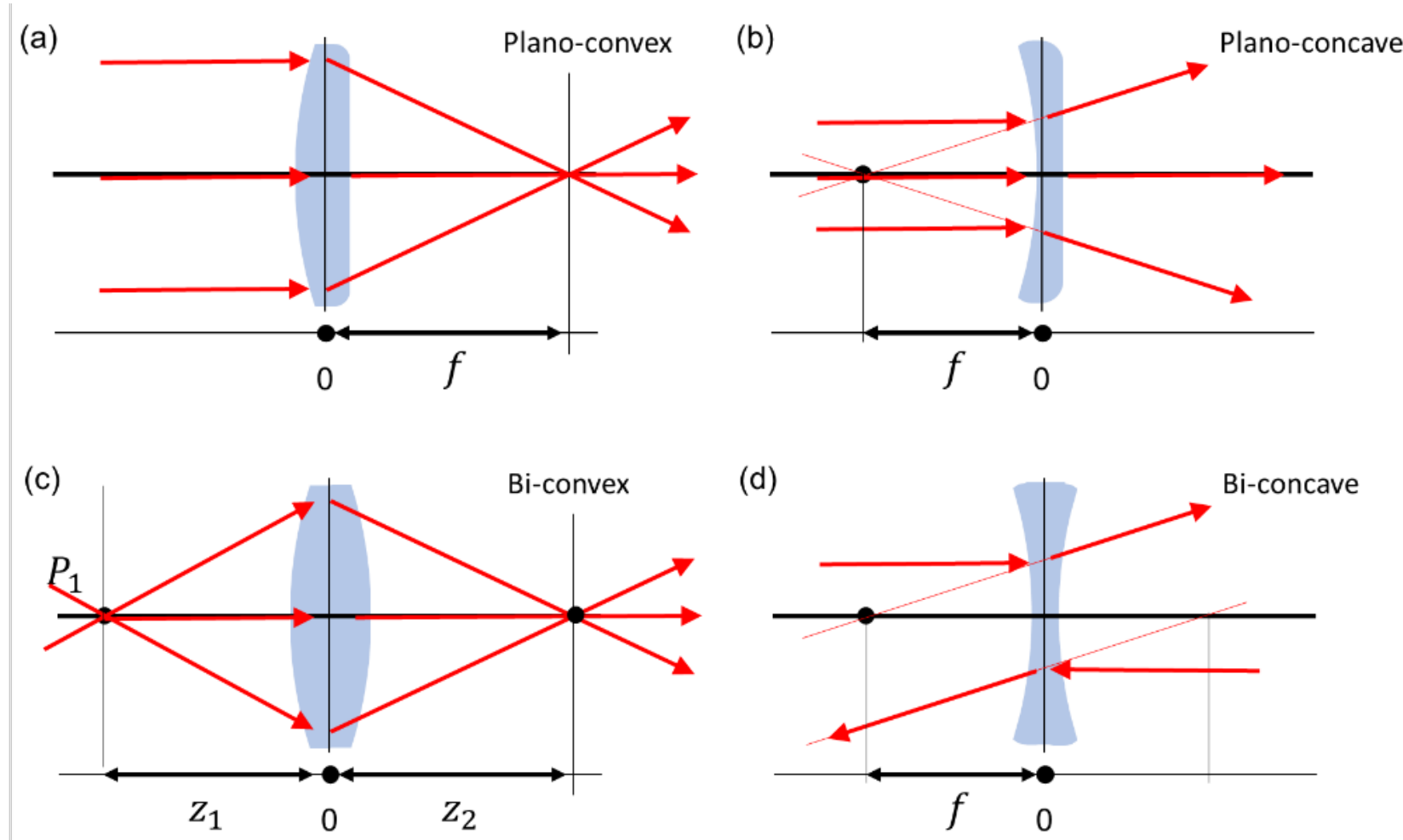


Lenses – Orientation to Infinity

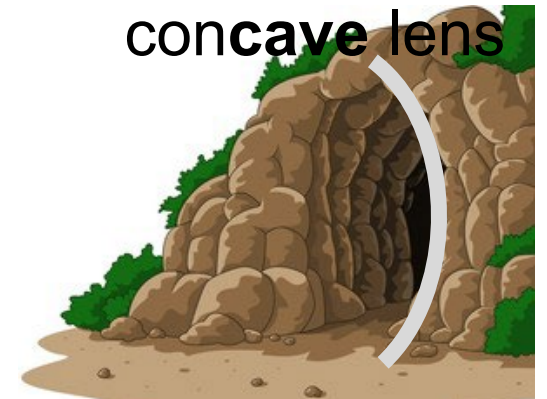


**Great tool for
visualization:**
<https://ricktu288.github.io/ray-optics/simulator/>

Lenses – Shapes

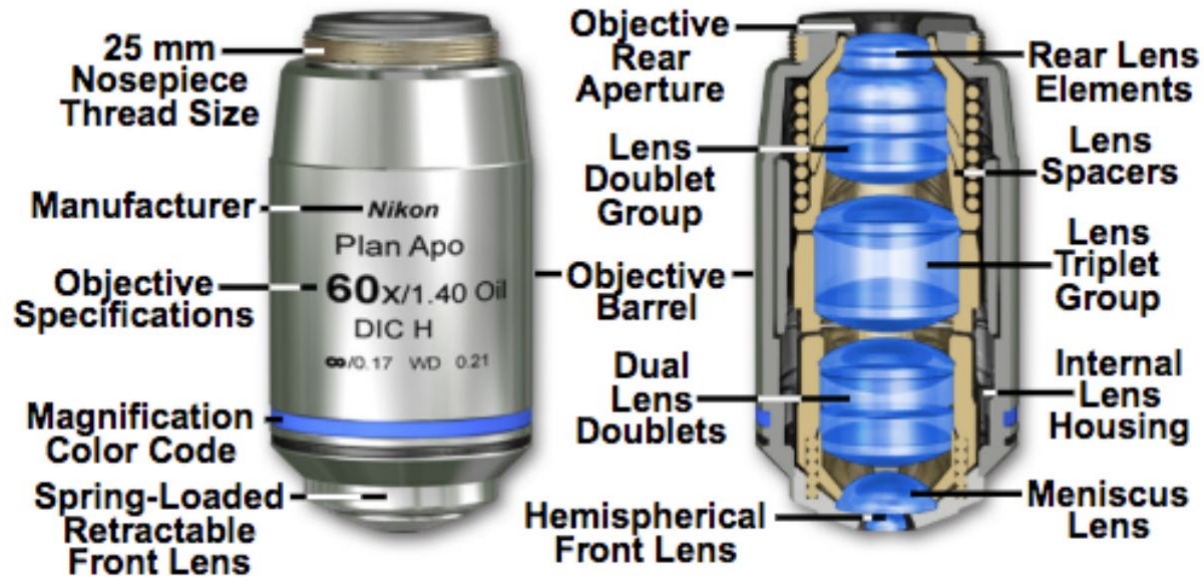


Tip to
remember:
concave lens

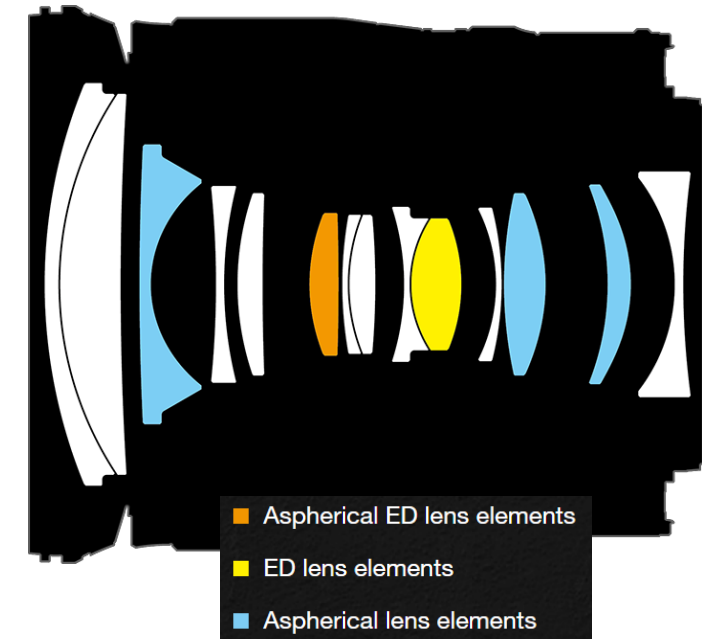


Lenses as assemblies

Lenses grouped together to correct for aberrations



<https://www.microscopyu.com/microscopy-basics/infinity-optical-systems>



http://www.nikon-asia.com/Zseries/en/nikkor-z-24-70mm-f-4-s_z6.html

Mirror

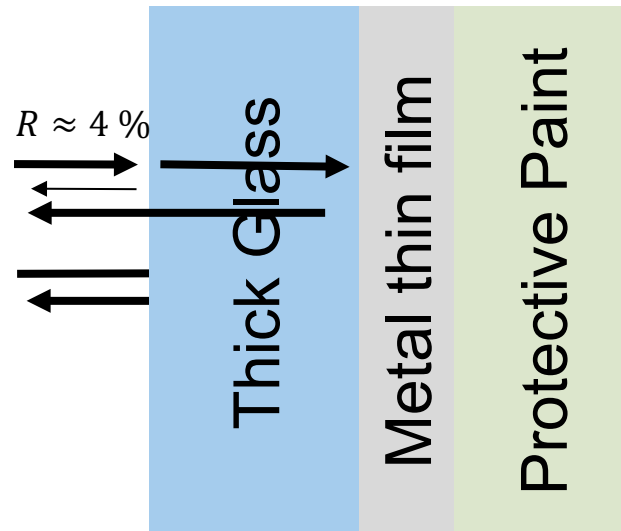
Household Mirror



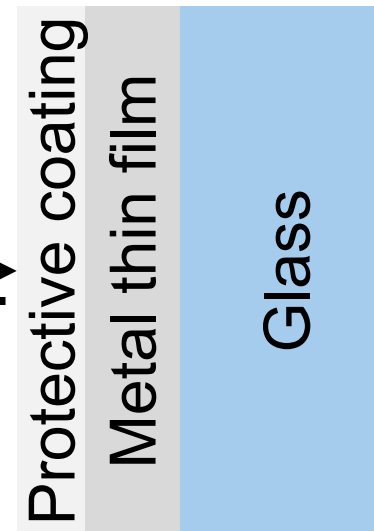
Metallic Mirror



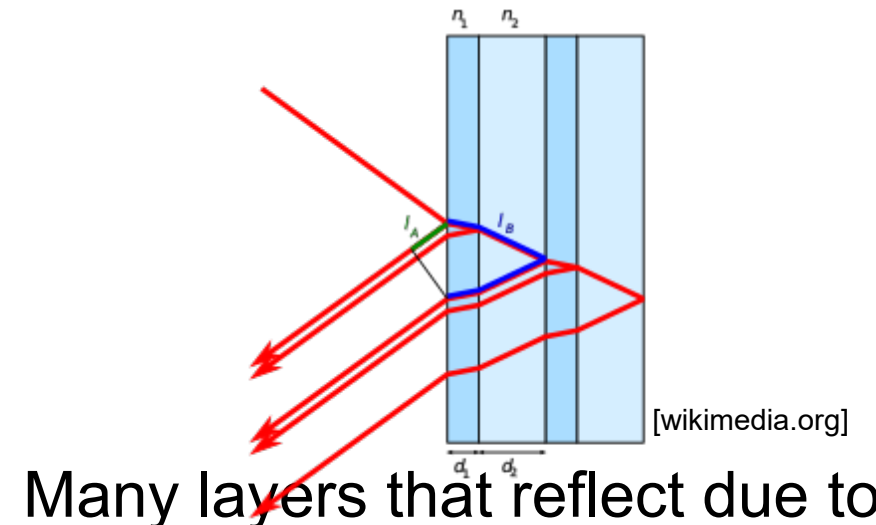
Dielectric / Bragg Mirror



Metallized glass
very smooth surface



Optional protective
Layer $\ll \lambda$

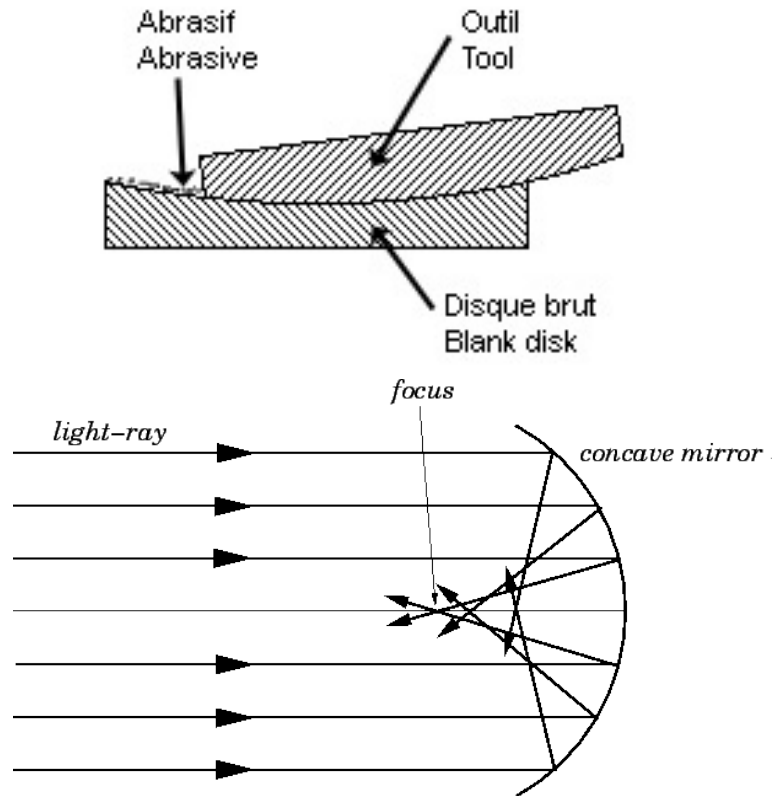


Many layers that reflect due to
constructive interference
(opposite of an antireflective
coating)

Mirrors – Shapes

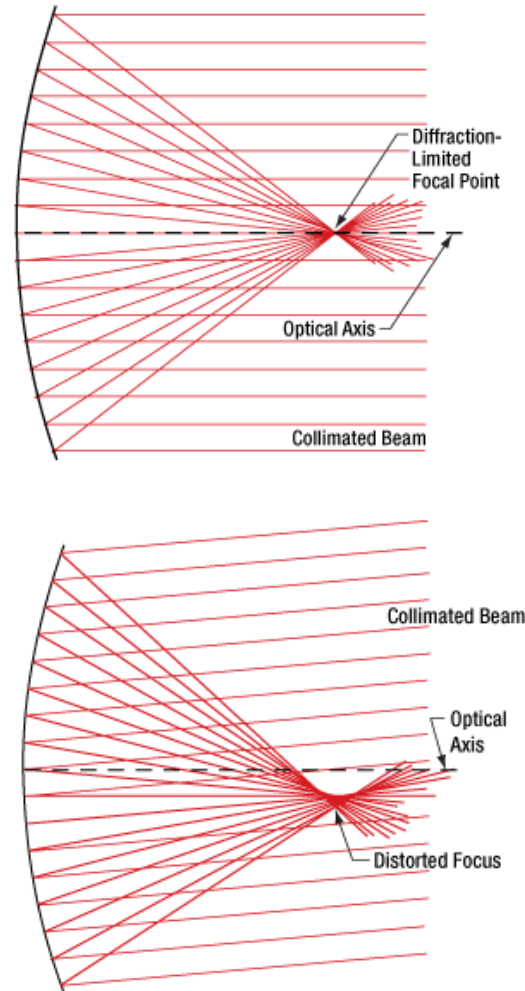
Spherical

easiest shape to fabricate



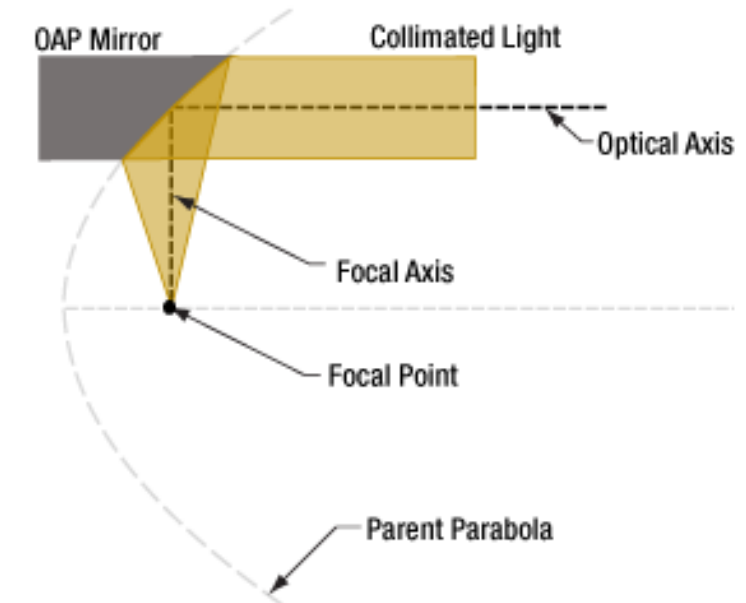
Parabolic

perfect focus point on-axis, coma off-axis



Off-axis parabolic

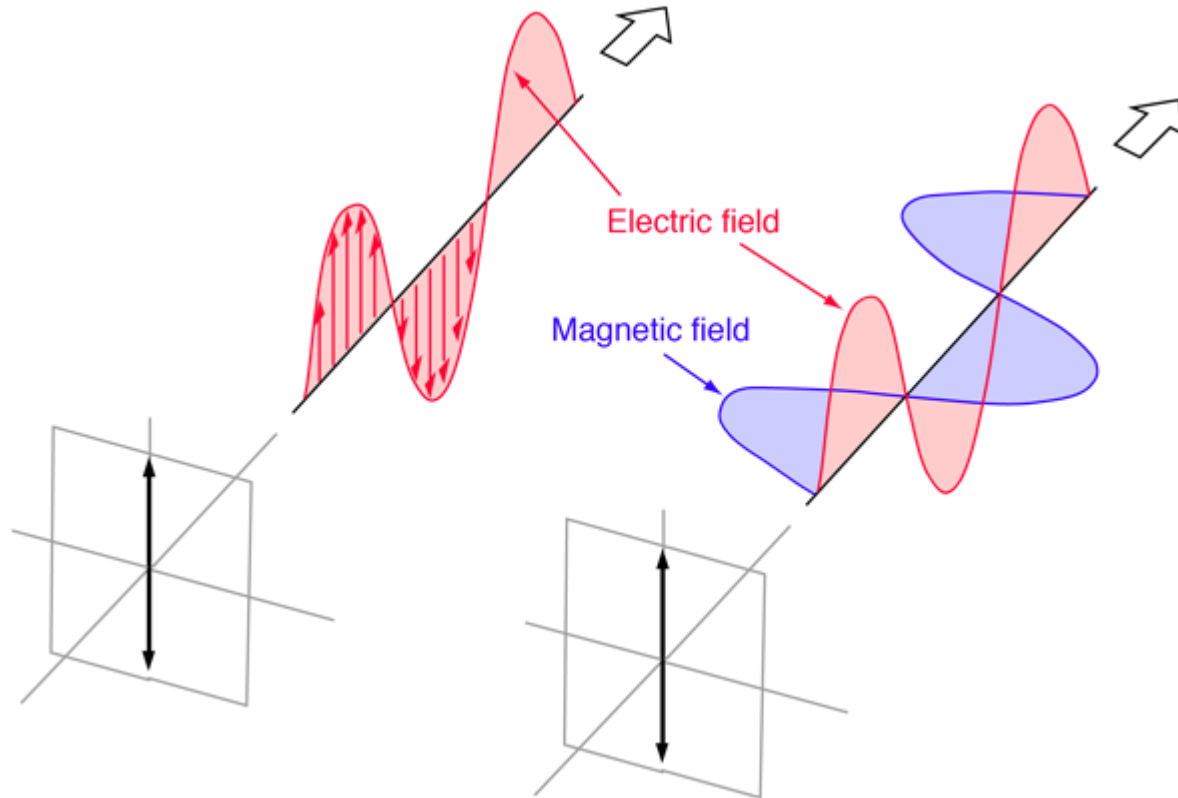
only part of a parabola



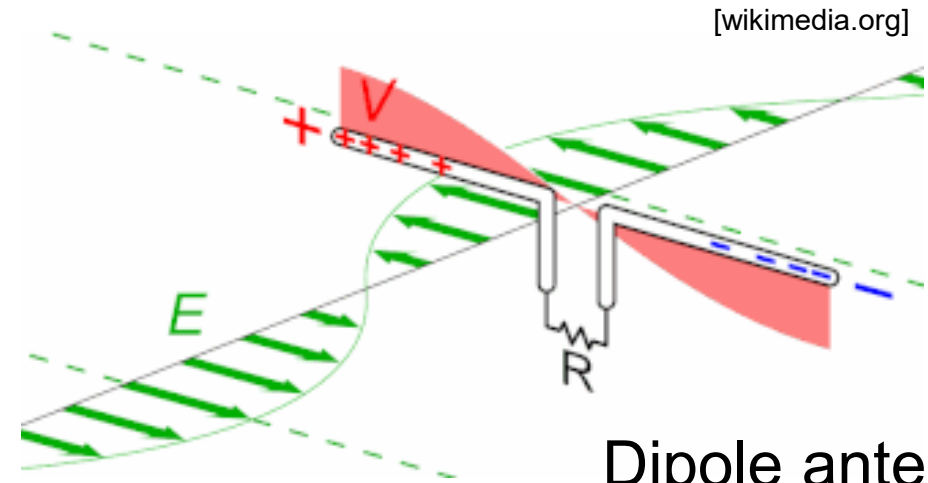
[<http://farside.ph.utexas.edu/teaching/302/lectures/node136.html>]

[thorlabs.com]

Polarization – linear

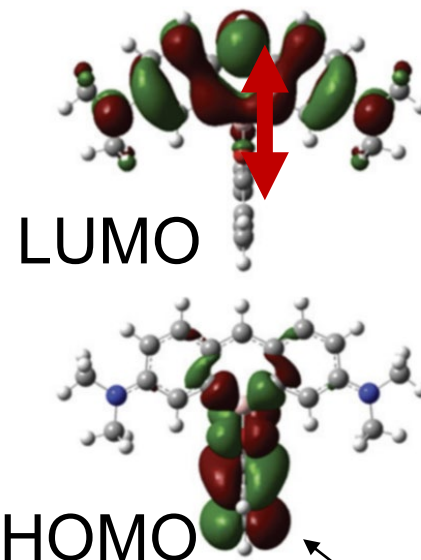
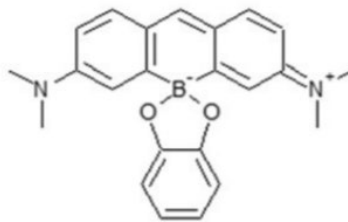


[<http://hyperphysics.phy-astr.gsu.edu/hbase/phyopt/polclas.html>]



Dipole antenna for
RF ($\sim$ MHz - GHz)

Red
fluorophore
JS-R/CA

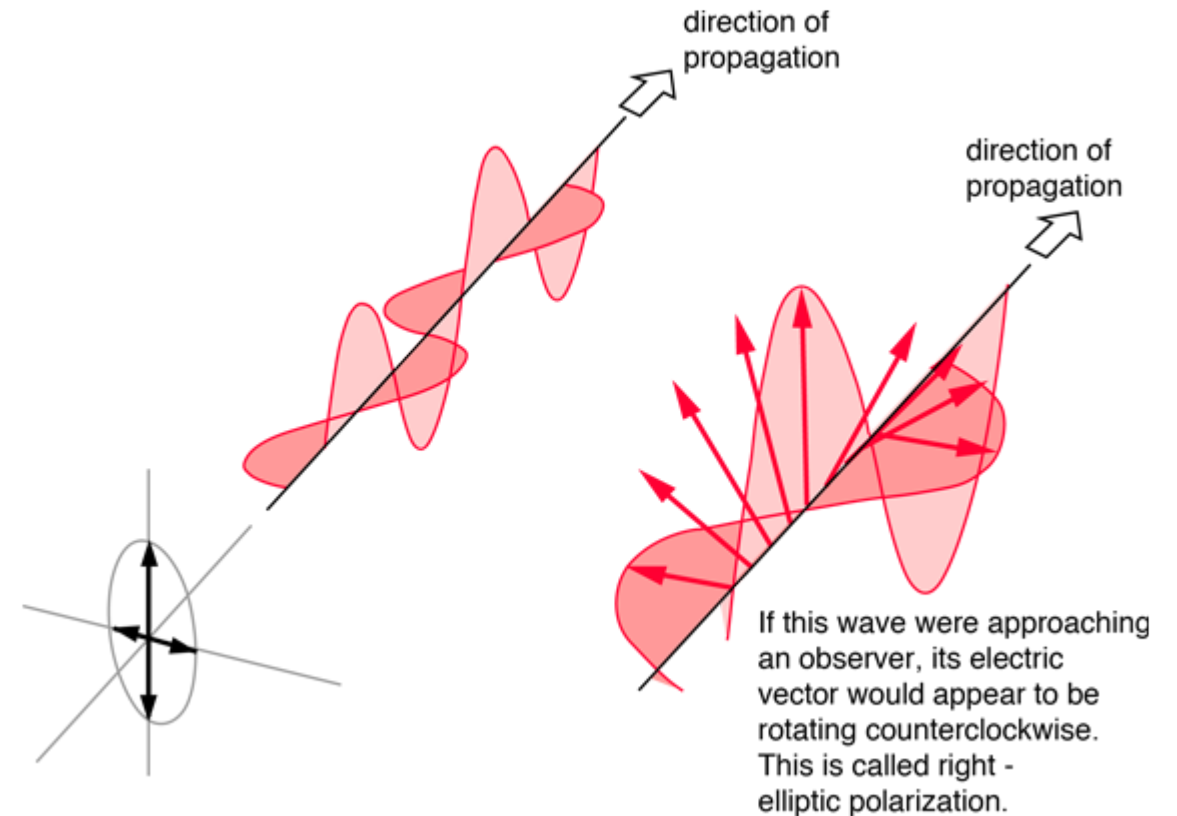
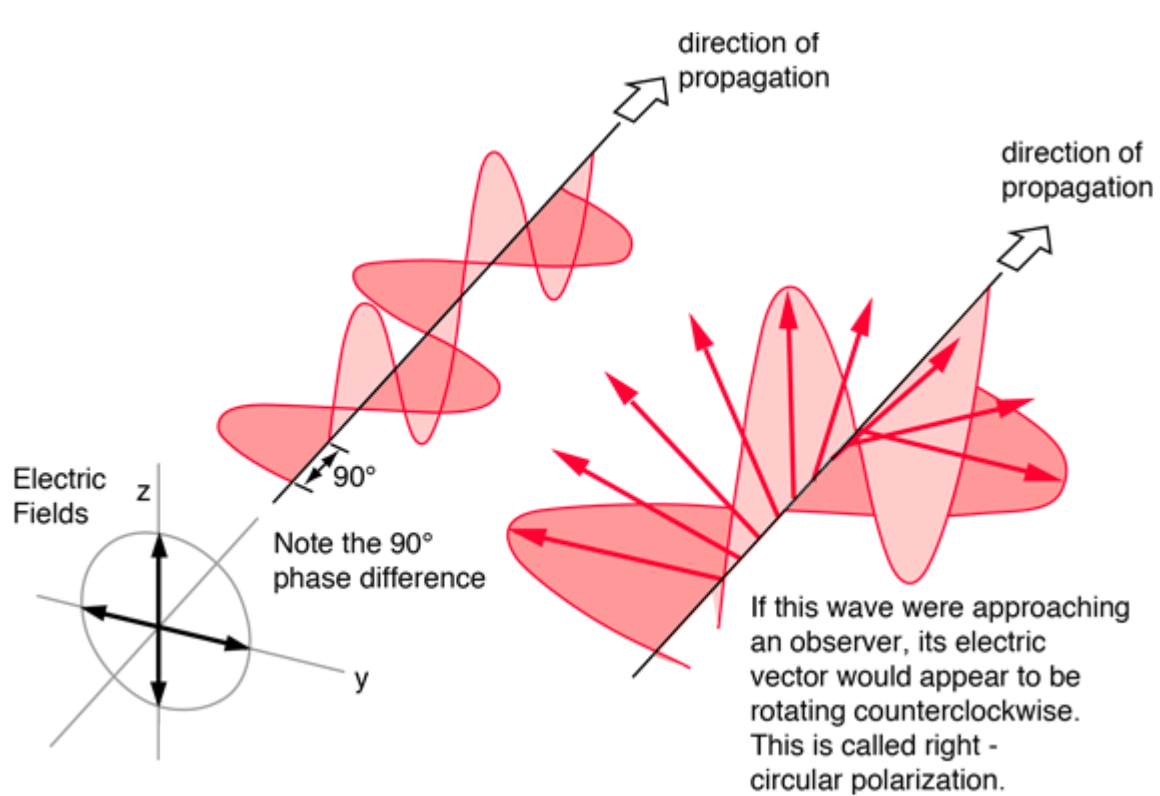


Dipole like
transition,
which emits
red light
($\sim$ 500 THz)

electron clouds

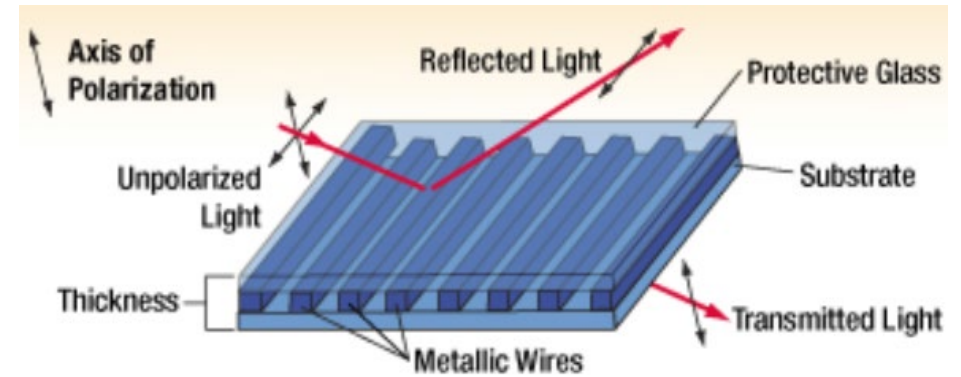
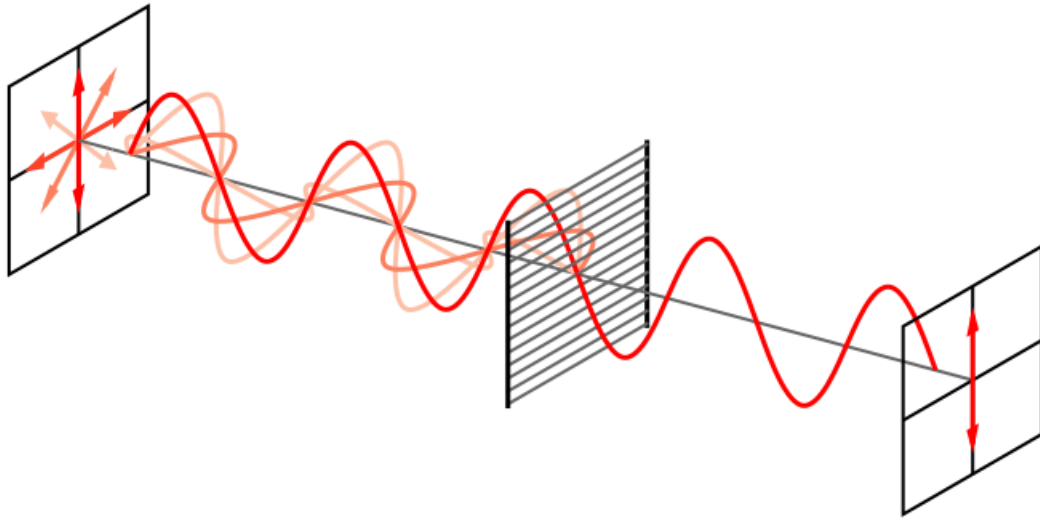
Shimomura, N et. al. 2016. A red fluorophore comprising a borinate-containing xanthene analogue as a polyol sensor. *Organic & biomolecular chemistry*, 14(42)

Polarization – Circular and Elliptical



[<http://hyperphysics.phy-astr.gsu.edu/hbase/phyopt/polclas.html>]

Linear Polarizers

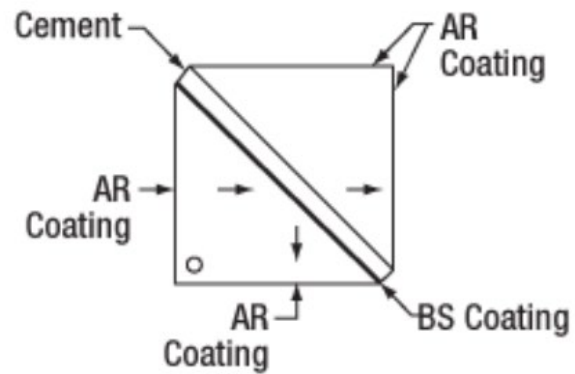


Made with:

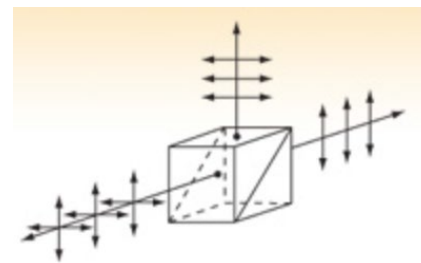
- Nanoscale metallic wires
- With iodine doped polyvinyl alcohol (PVA) sheets with ordered orientation (used in Polaroid glasses, LCD screens)
- Birefringent crystals

Beamsplitters

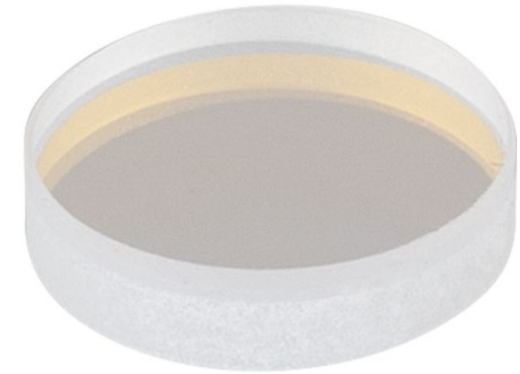
50:50, 90:10, ... Splitters



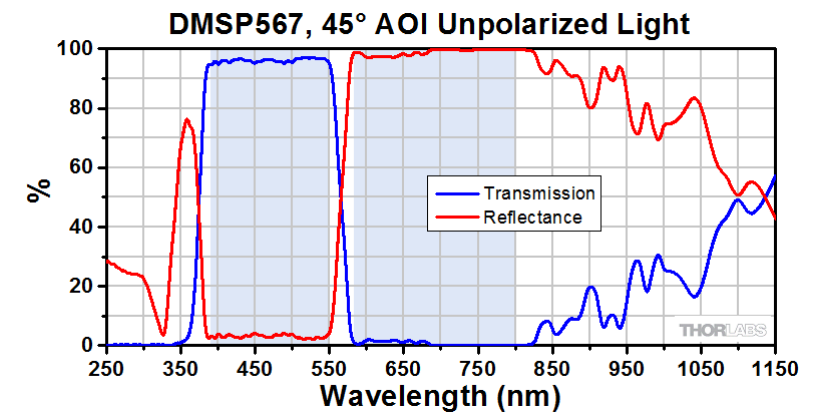
Polarizing Beamsplitters

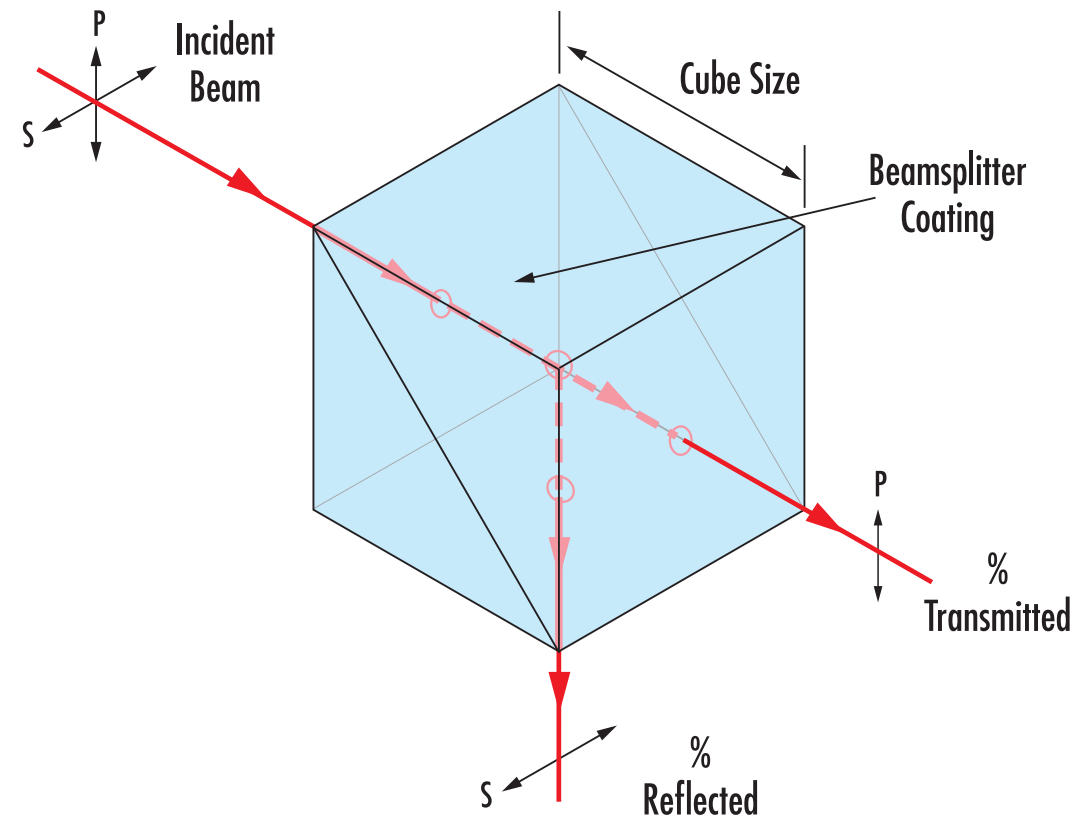
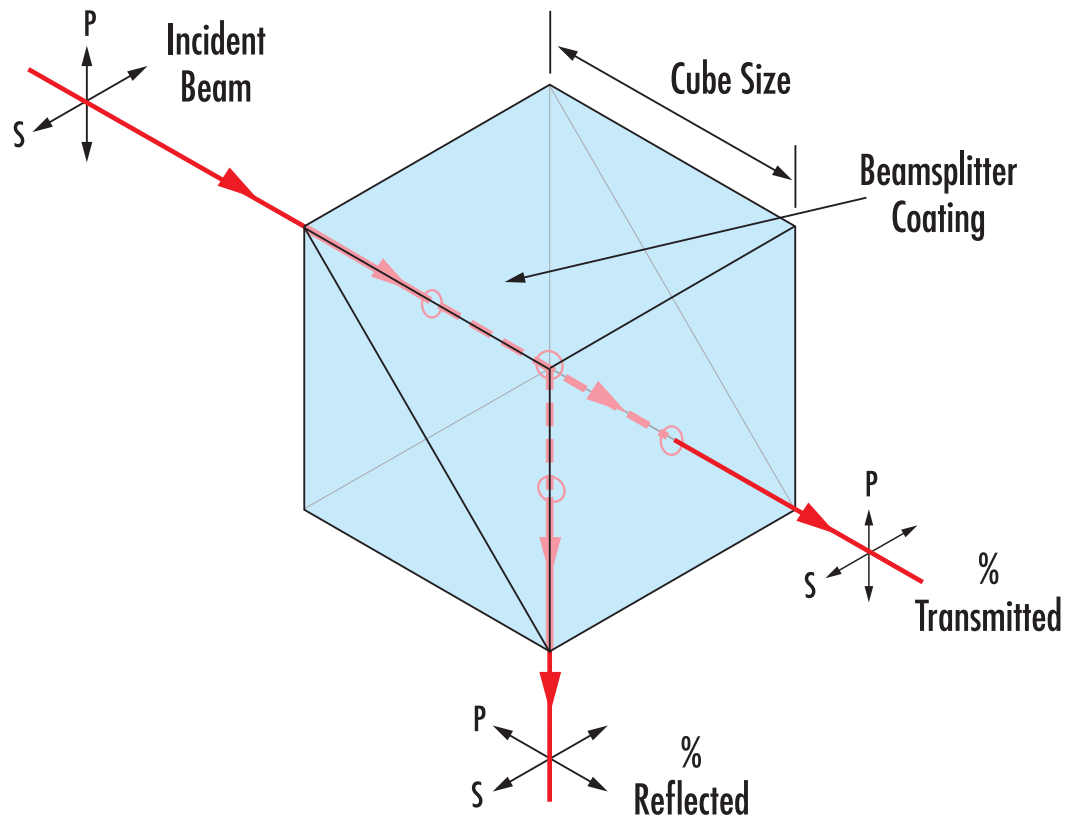


Dichroic Splitters



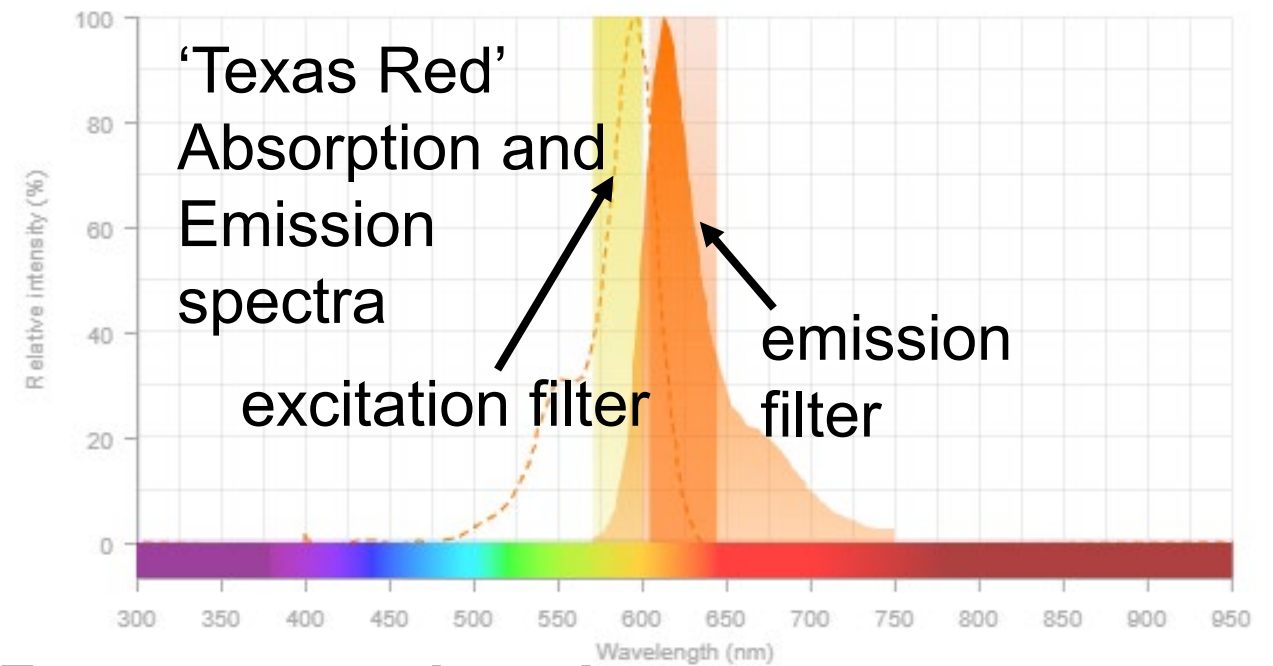
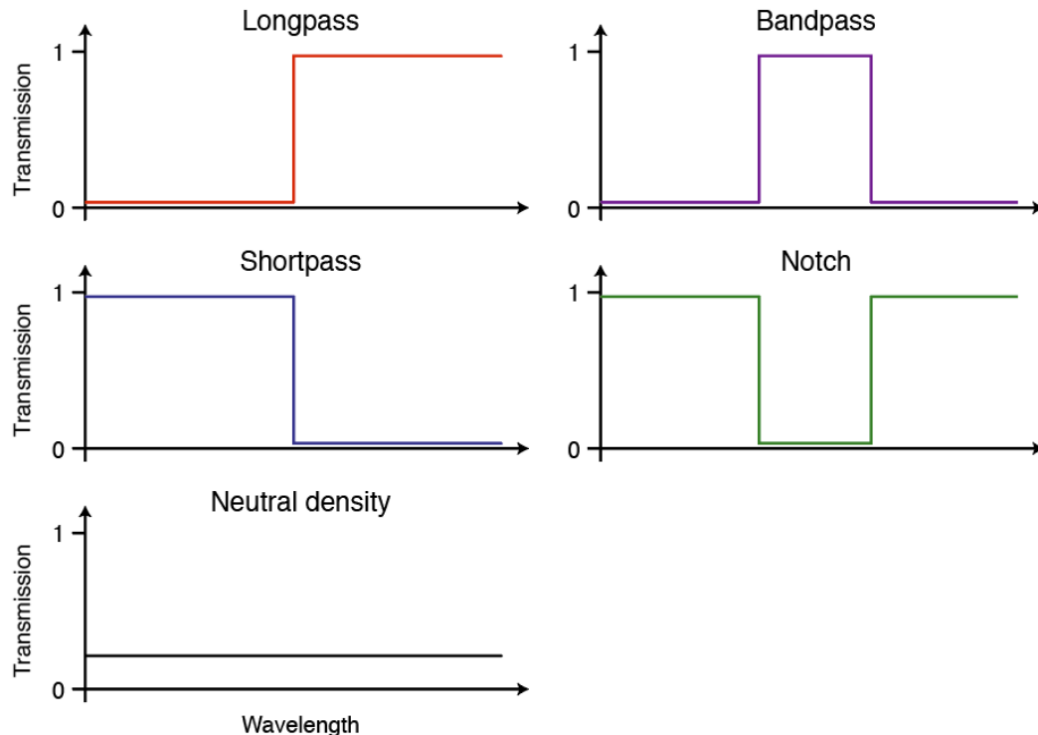
DMSP567T





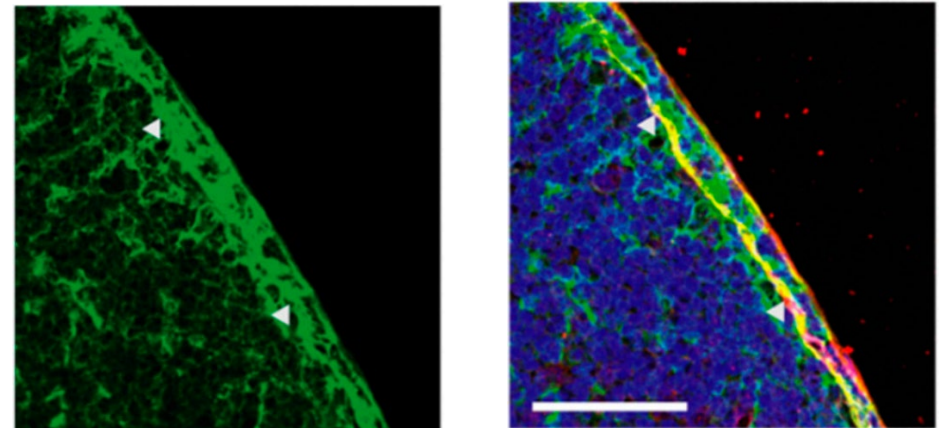
Filters

- Wavelength filters can be made absorptive or reflective
- Most often dielectric interference filters



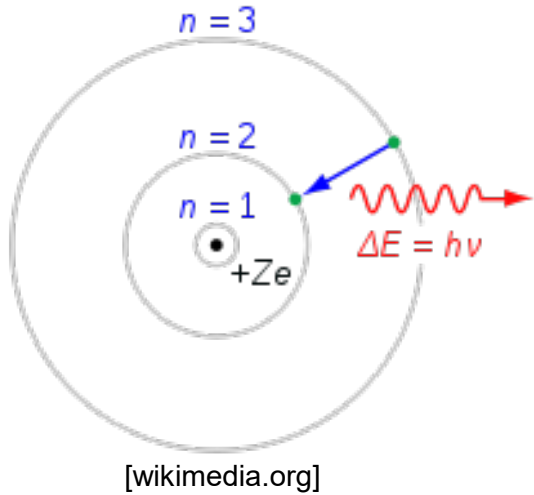
Fluorescence imaging

Label specific parts of a cell with a fluorescent marker

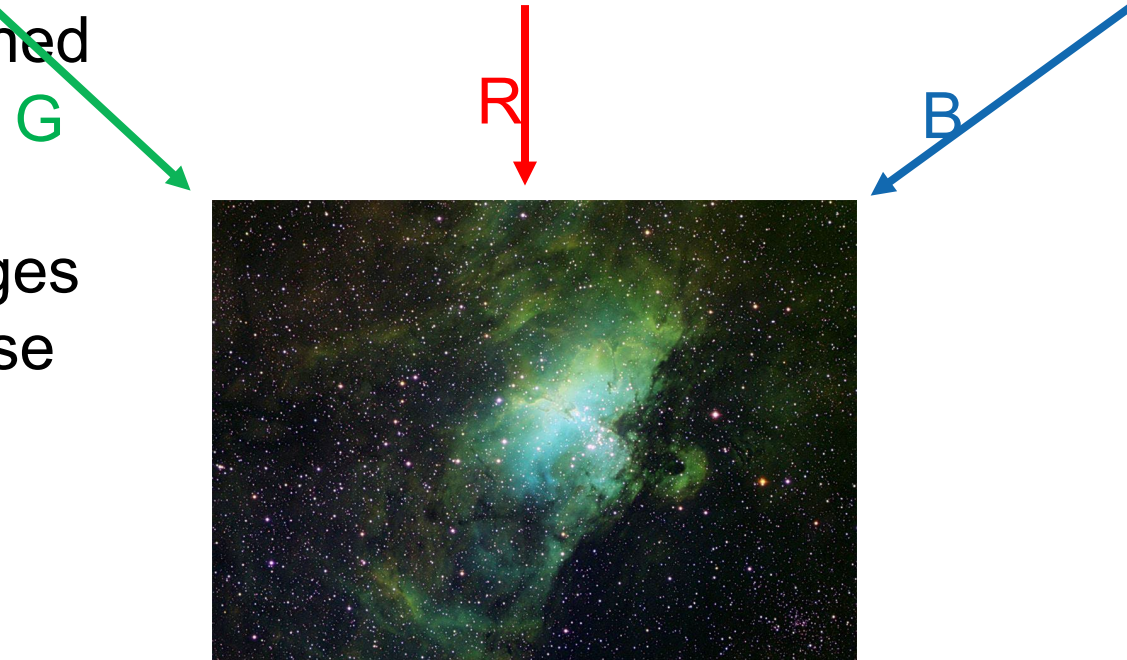


Sibler, E. *et al.* Single-Cell Transcriptional Heterogeneity of Lymphatic Endothelial Cells in Normal and Inflamed Murine Lymph Nodes. *Cells* **10**, 1371 (2021)

Narrowband Filters in Astronomy



- Atomic emission lines are well defined
e.g. $\lambda_{H\alpha} = 656 \text{ nm}$
- Telescopes (e.g. Hubble) take images with very narrowband filters for these lines
- Rejects any stray light (noise) at different wavelengths

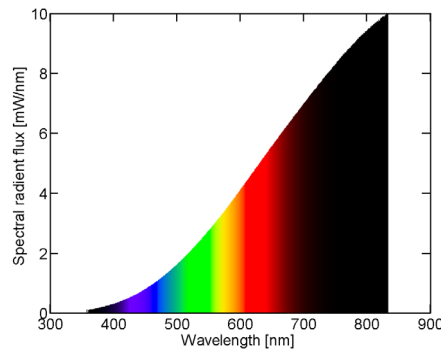


Light sources

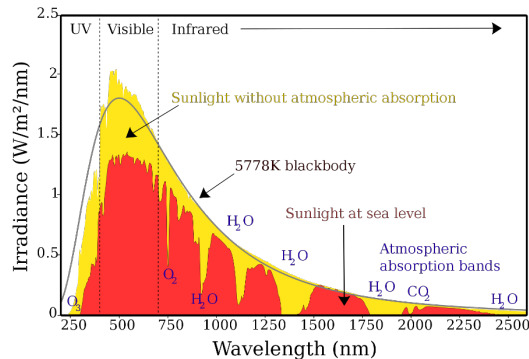
Incandescent Lights

Hot filament

Blackbody radiation



Spectrum of Solar Radiation (Earth)



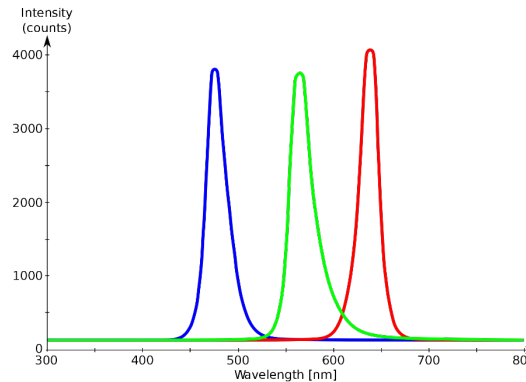
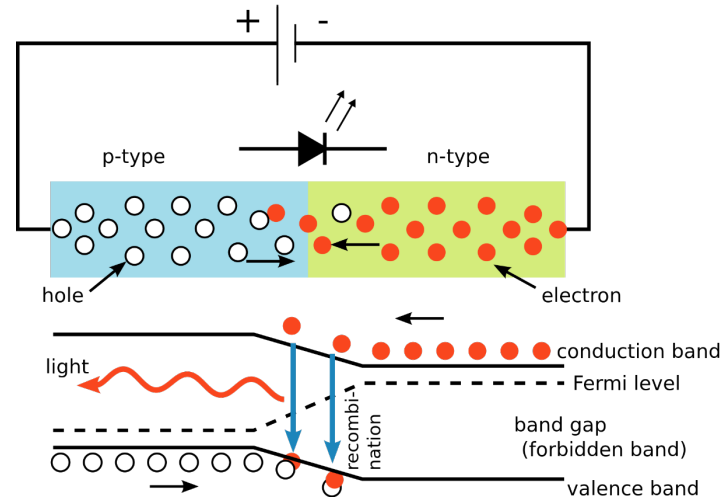
Light Emitting Diodes

Radiative

recombination of

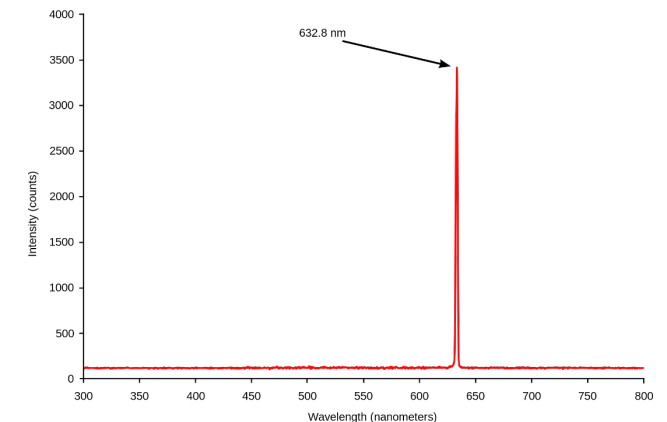
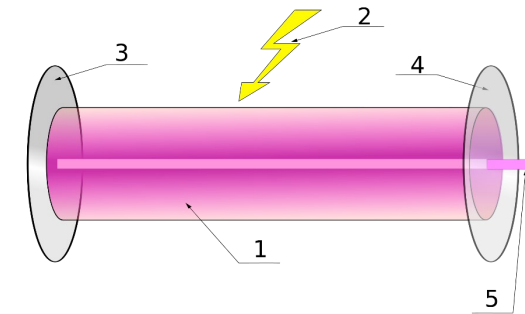
charge carriers inside

a semiconductor



Light Amplification by Stimulated Emission of Radiation – LASER

Stimulated emission of radiation inside a gain medium with optical feedback



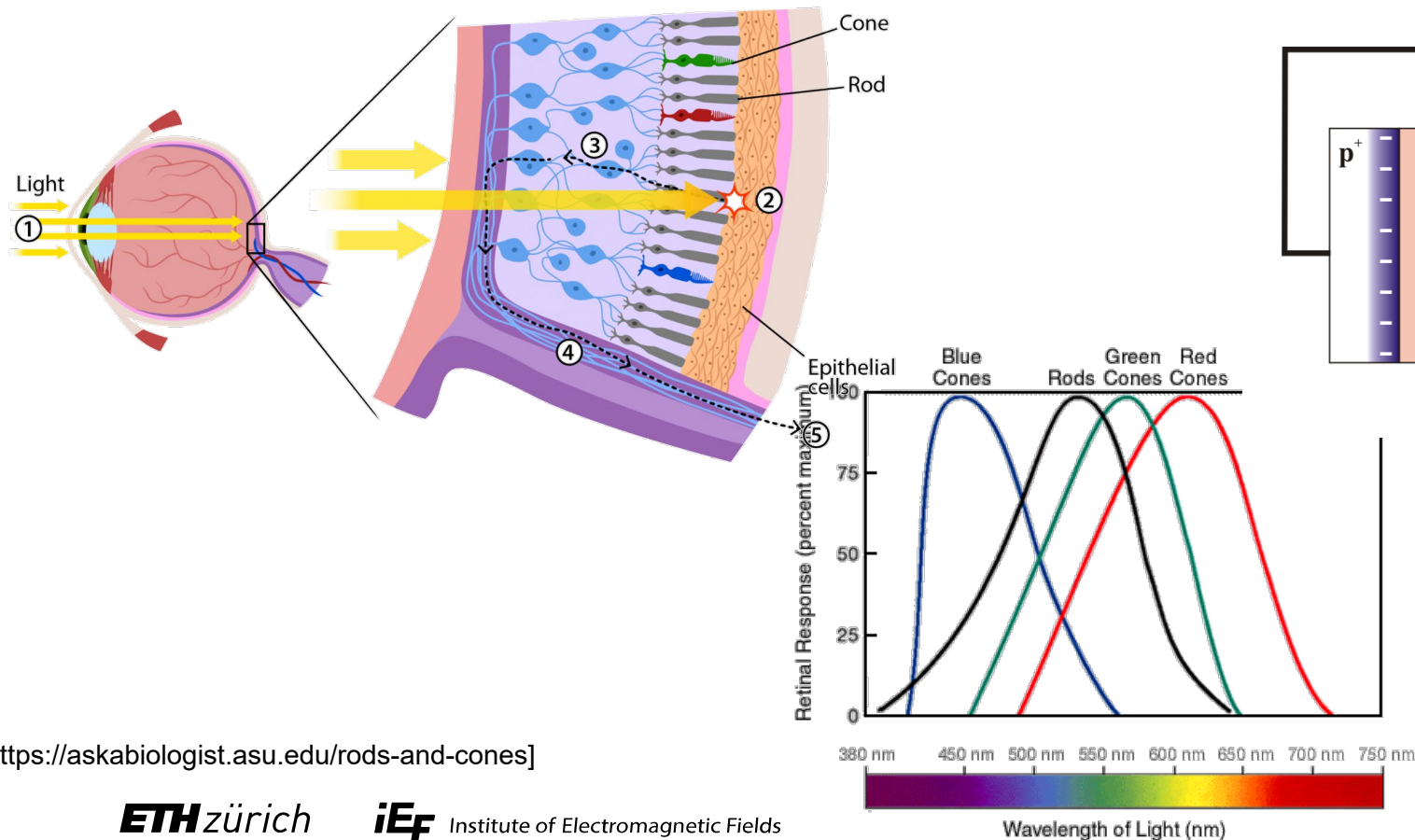
Detectors

Human Eye

Comes with a free lens

4 different detecting elements (RGB + Intensity)

$$\lambda_{\text{detection}} \approx 400 - 700 \text{ nm}$$



[<https://askabiologist.asu.edu/rods-and-cones>]

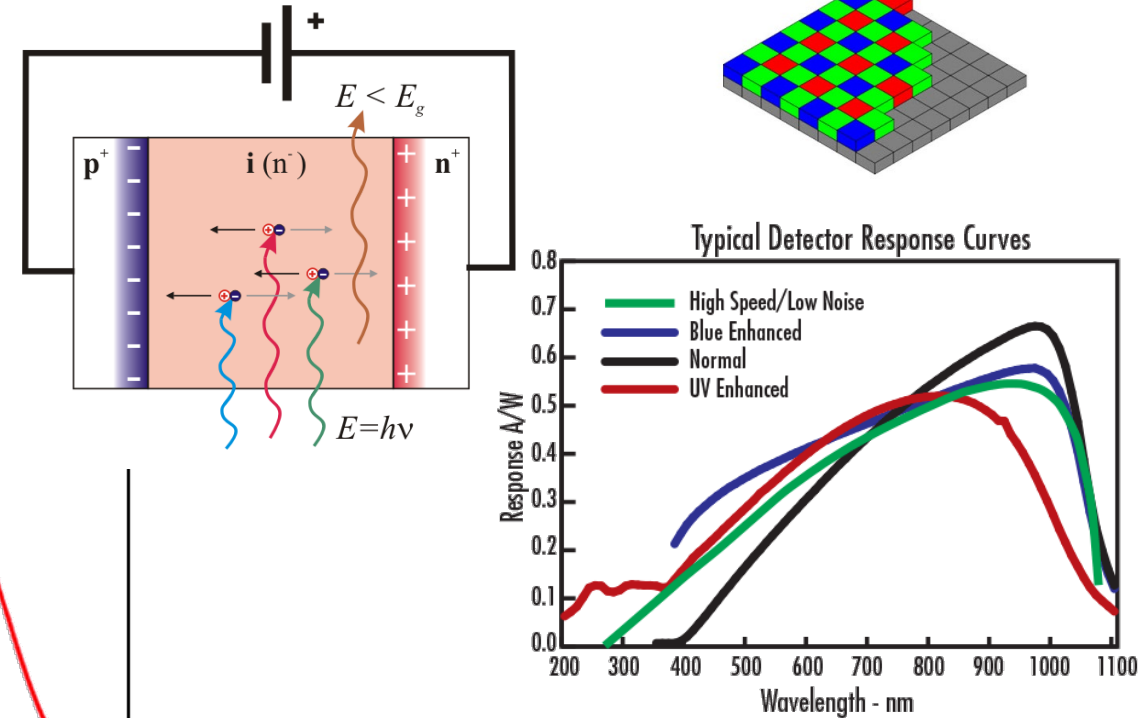
Semiconductor Silicon

No free lens

Color filtering separate

$$\lambda_{\text{detection,Si}} \approx 200 - 1100 \text{ nm}$$

Other Semiconductors possible (Ge, InGaAs, HgCdTe)



[<https://physicsopenlab.org/2017/04/28/si-pin-photodiode-%CE%B2-detector/>]
[edmundoptics.com]

